汽车电子与控制
Automotive Electronics and Control Systems (AECS)

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2021.10

第二章：汽车电控系统数字核心（单片机）
Digital Core of AECS (Microcontroller)

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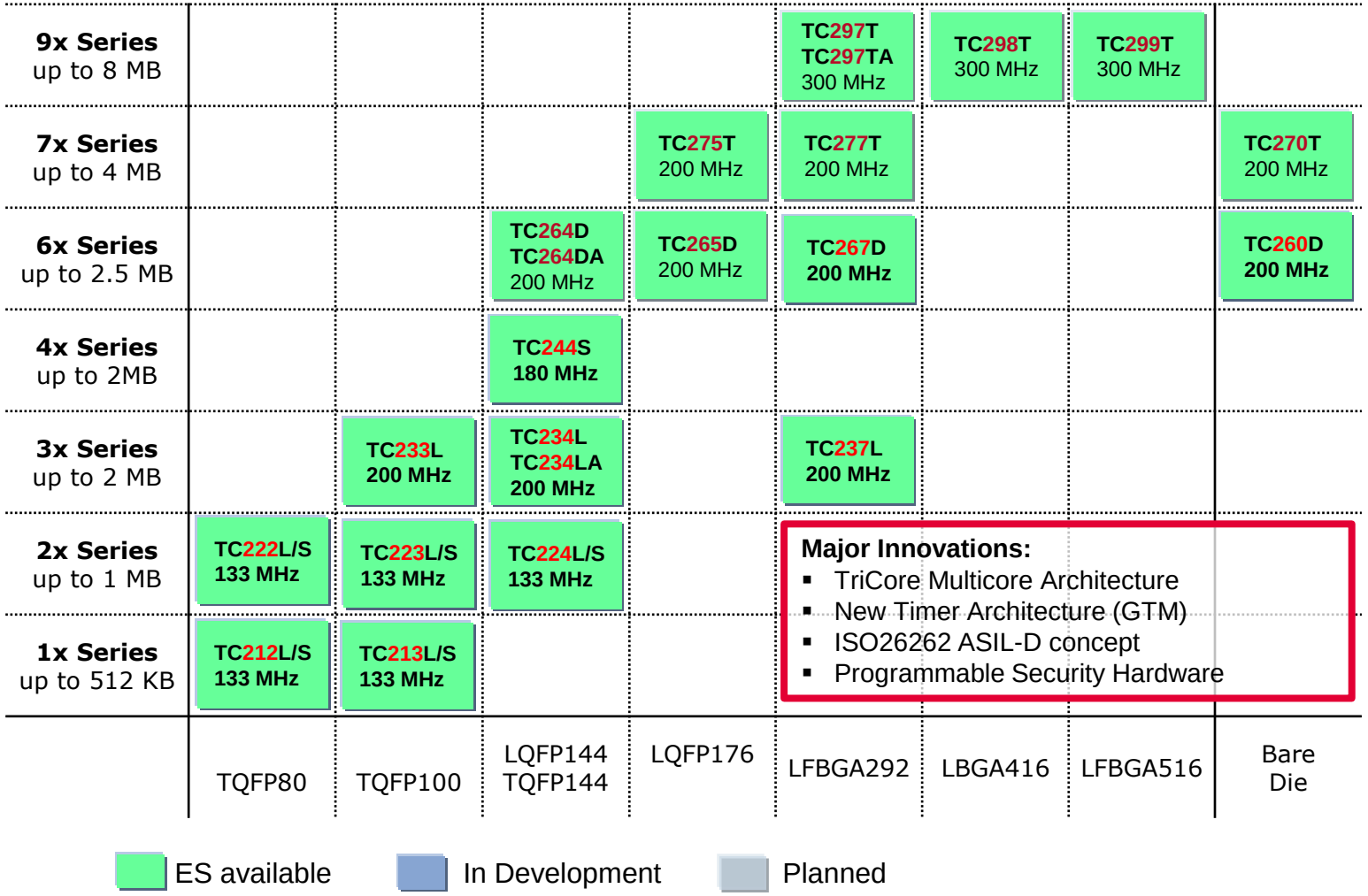
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3 TC系列MCU的CPU与外围电路

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- （2）CPU架构、主要寄存器及工作原理
- （3）MCU内存管理

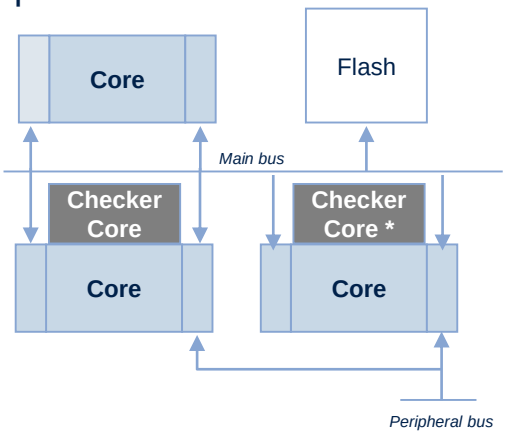

9x Series
up to 8 MB
TC297T
TC297TA
300 MHz
TC298T
300 MHz
TC299T
300 MHz
7x Series
up to 4 MB
TC275T
200 MHz
TC277T
200 MHz
TC270T
200 MHz
6x Series
up to 2.5 MB
TC264D
TC264DA
200 MHz
TC265D
200 MHz
TC267D
200 MHz
TC260D
200 MHz
4x Series
up to 2MB
TC244S
180 MHz
3x Series
up to 2 MB
TC233L
200 MHz
TC234L
TC234LA
200 MHz
TC237L
200 MHz
2x Series
up to 1 MB
TC222L/S
133 MHz
TC223L/S
133 MHz
TC224L/S
133 MHz
1x Series
up to 512 KB
TC212L/S
133 MHz
TC213L/S
133 MHz
Major Innovations:
TriCore Multicore Architecture
New Timer Architecture (GTM)
ISO26262 ASIL-D concept
Programmable Security Hardware
TQFP80
TQFP100
LQFP144
TQFP144
LQFP176
LFBGA292
LBGA416
LFBGA516
Bare
Die
ES available
In Development
Planned

AURIX Multi-Core Architecture

Triple Core Lockstep

-

„**T**“ Marking

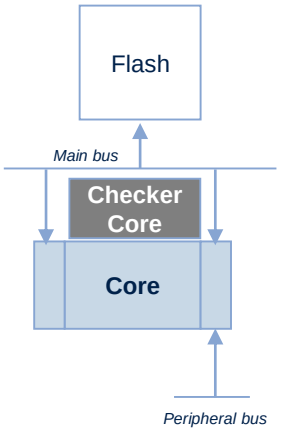


TC29xT and TC27xT

Lockstep

-

„**L**“ Marking

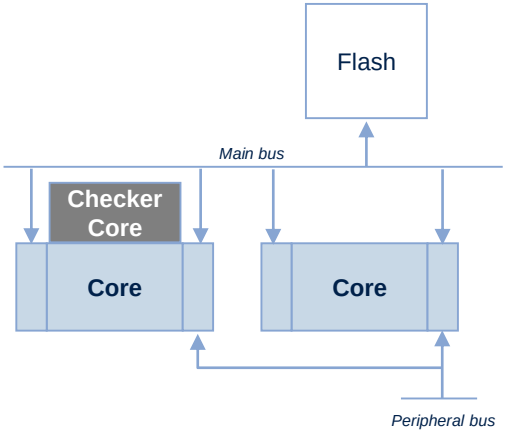


TC21xL,TC22xL and TC23xL

Dual Core Lockstep

-

„**D**“ Marking

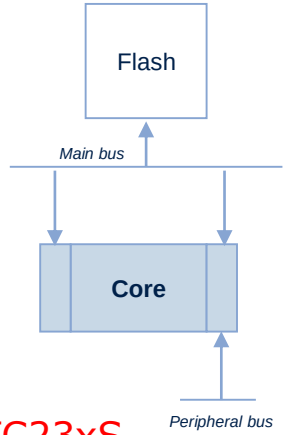


TC26xD

Single Core

-

„**S**“ Marking



TC21xS,TC22xS and TC23xS

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AURIX Multi-Core Architecture

SEooC Specification:

Safe Element out of Context

Safety Architecture

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1.1 System Architecture of the TC21x/TC22x/TC23x

The TC21x/TC22x/TC23x combines three powerful technologies within one silicon die, achieving new levels of power, speed, and economy for embedded applications:

- Reduced Instruction Set Computing (RISC) processor architecture
- Digital Signal Processing (DSP) operations and addressing modes
- On-chip memories and peripherals

DSP operations and addressing modes provide the computational power necessary to efficiently analyse complex real-world signals. The RISC load/store architecture provides high computational bandwidth with low system cost. On-chip memory and peripherals are designed to support even the most demanding high-bandwidth real-time embedded control-systems tasks.

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System Architecture of the TC21x/TC22x/TC23x

Additional high-level features of the TC21x/TC22x/TC23x include:

- Efficient memory organization: instruction and data scratch memories, caches
- Serial communication interfaces – flexible synchronous and asynchronous modes
- DMA Controller – DMA operations and interrupt servicing
- Flexible interrupt system – configurable interrupt priorities and targets
- General-purpose timers
- High-performance on-chip buses
- On-chip debugging and emulation facilities
- Flexible interconnections to external components
- Flexible power-management

DMA：直接内存访问
Direct Memory Access

The TC21x/TC22x/TC23x is a high-performance microcontroller with one TriCore CPU, program and data memories, buses, bus arbitration, interrupt system, DMA controller and a powerful set of on-chip peripherals. The TC21x/TC22x/TC23x is designed to meet the needs of the most demanding embedded control systems applications where the competing issues of price/performance, real-time responsiveness, computational power, data bandwidth, and power consumption are key design elements.

The TC21x/TC22x/TC23x offers several versatile on-chip peripheral units such as serial controllers, timer units, and analog-to-digital converters. Within the TC21x/TC22x/TC23x, all these peripheral units are connected to the TriCore CPU /

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System Architecture of the TC21x/TC22x/TC23x

Table 1 Overview of TC21x/TC22x/TC23x Family Devices

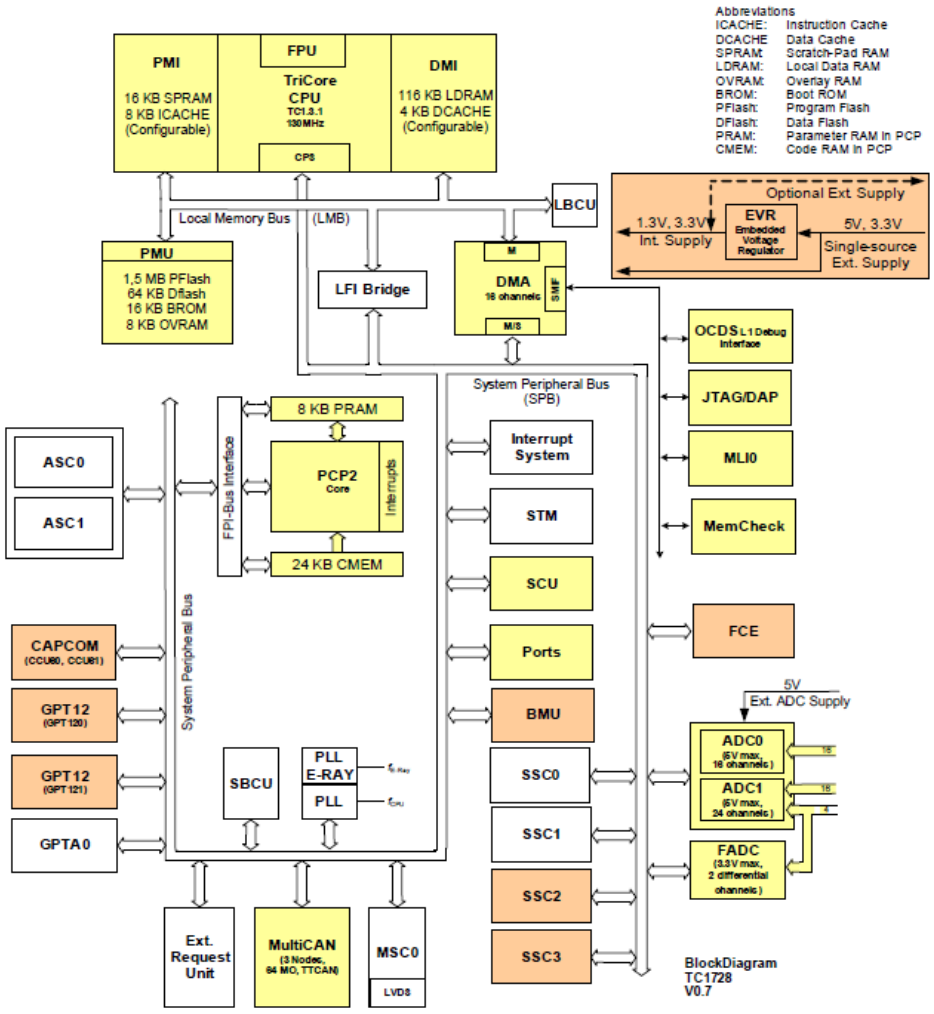
Feature		TC21x	TC22x	TC23x	TC23x ADAS
CPU Core	Type	TC1.6E			
	Cores / Checker Cores	1 / 1[1)]		1 / 1	
	Max. Freq.	133 MHz		200 MHz	
	FPU	yes			
	MPU	yes			
Program Flash	Size	512 Kbyte	1 Mbyte	2 Mbyte	
Data Flash	Size	64 Kbyte	96 Kbyte	128 Kbyte	
Cache	Instruction	8 Kbyte			
	Data	4 line read buffer			
SRAM	Size (DSPR/PSPR)	48 / 8 Kbyte	88 / 8 Kbyte	184 / 8 Kbyte	
	Size LMU	0 Kbyte			32 Kbyte
	Size EMEM	0 Kbyte			512 Kbyte
DMA	Channels	16			
ADC (SAR)	Voltage range	0 … 5.5 V (determined by ADC supply)			
	Inputs	24			
	Converters	2			4
	Resolution	12 Bits			
	Channels	12 + 12			12 + 12 + overlay 1 + 1

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Architecture: Single / Multi-CPU

Parallel Access in Audo-family , Single CPU:

- **CPU instruction fetches from**
  - PFlash
  - External Memory
  - Program or Data Scratch Pad SRAM (PSPR / DSPR)
  - LMU SRAM
- **CPU data access**
  - Data Scratch Pad SRAM (DSPR)
  - System Peripheral Bus (SPB)
  - other SRAMs (LMU SRAM, PSPR)
- **PCP to SRAMs**
- **DMA access to SRAMs**



System Architecture of the TC21x/TC22x/TC23x

- Shared Resource Interconnect (SRI)
- System Peripheral Bus (SPB)

SRI: Shared Resource Interconnect

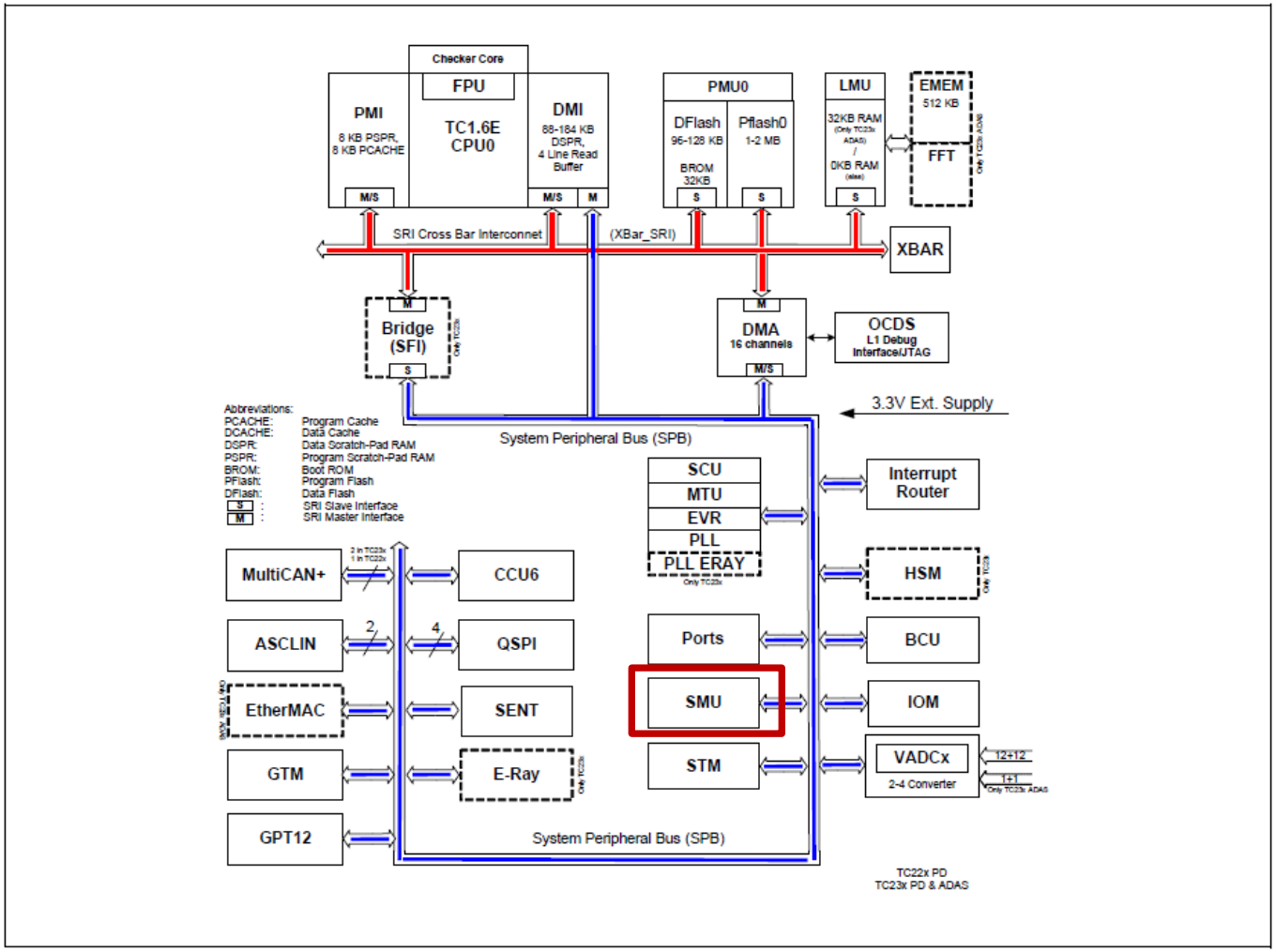
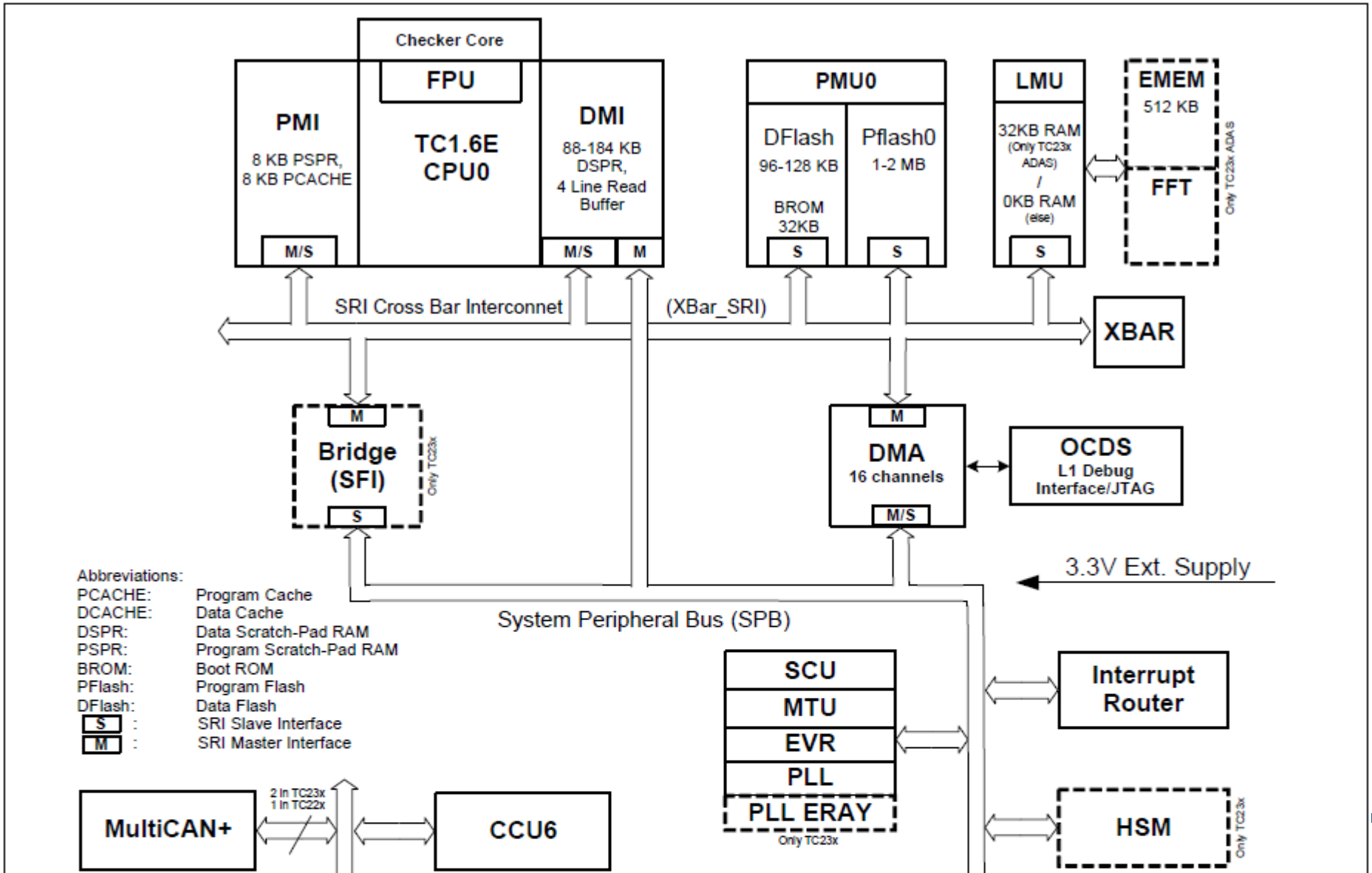


Figure 3-1 On Chip Buses in TC21x/TC22x/TC23x Processor Subsystem

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System Architecture of the TC21x/TC22x/TC23x



System Architecture of the TC21x/TC22x/TC23x

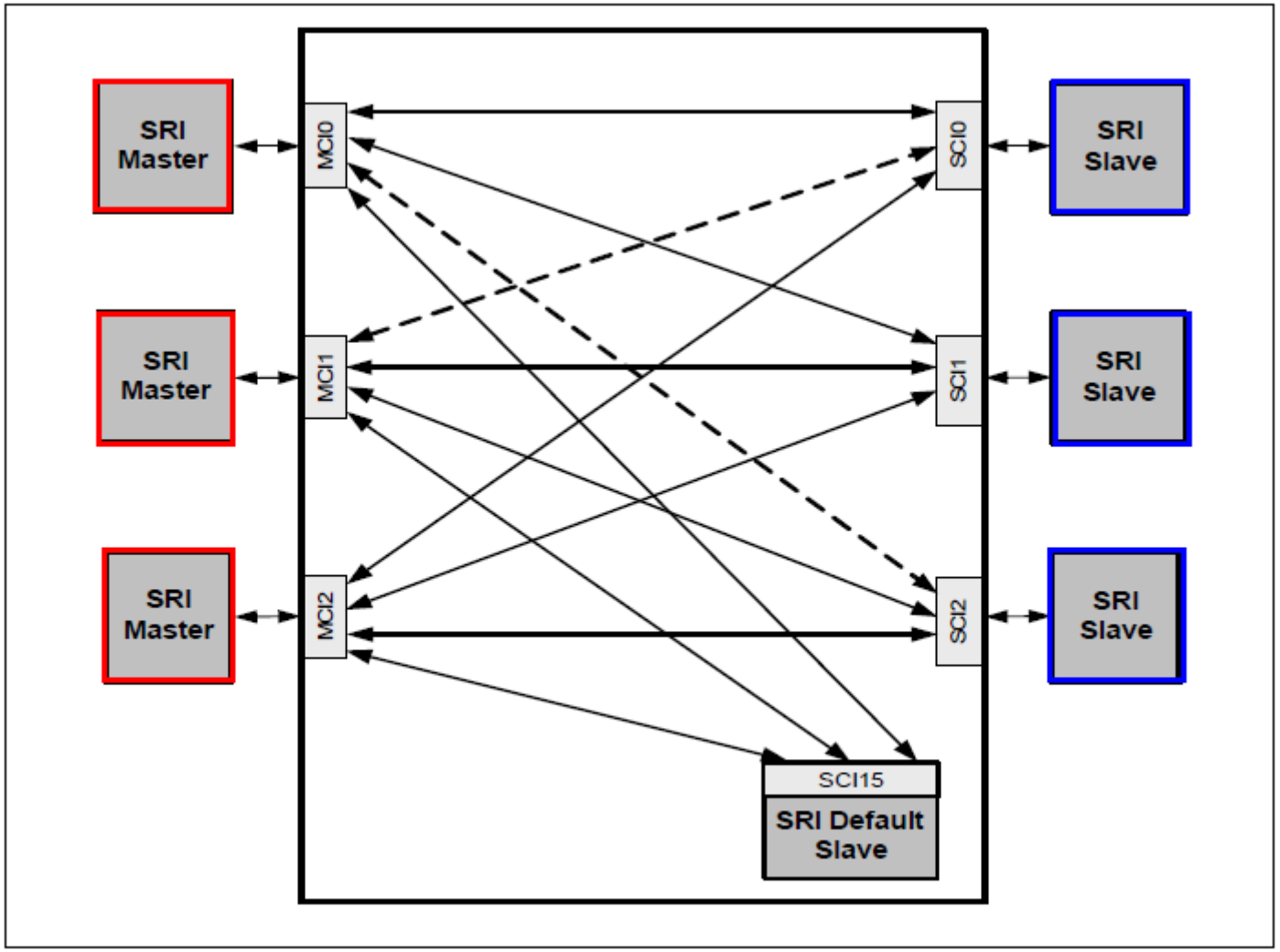


Figure 3-2 XBAR_SRI point to point connection scheme

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System Architecture of the TC21x/TC22x/TC23x

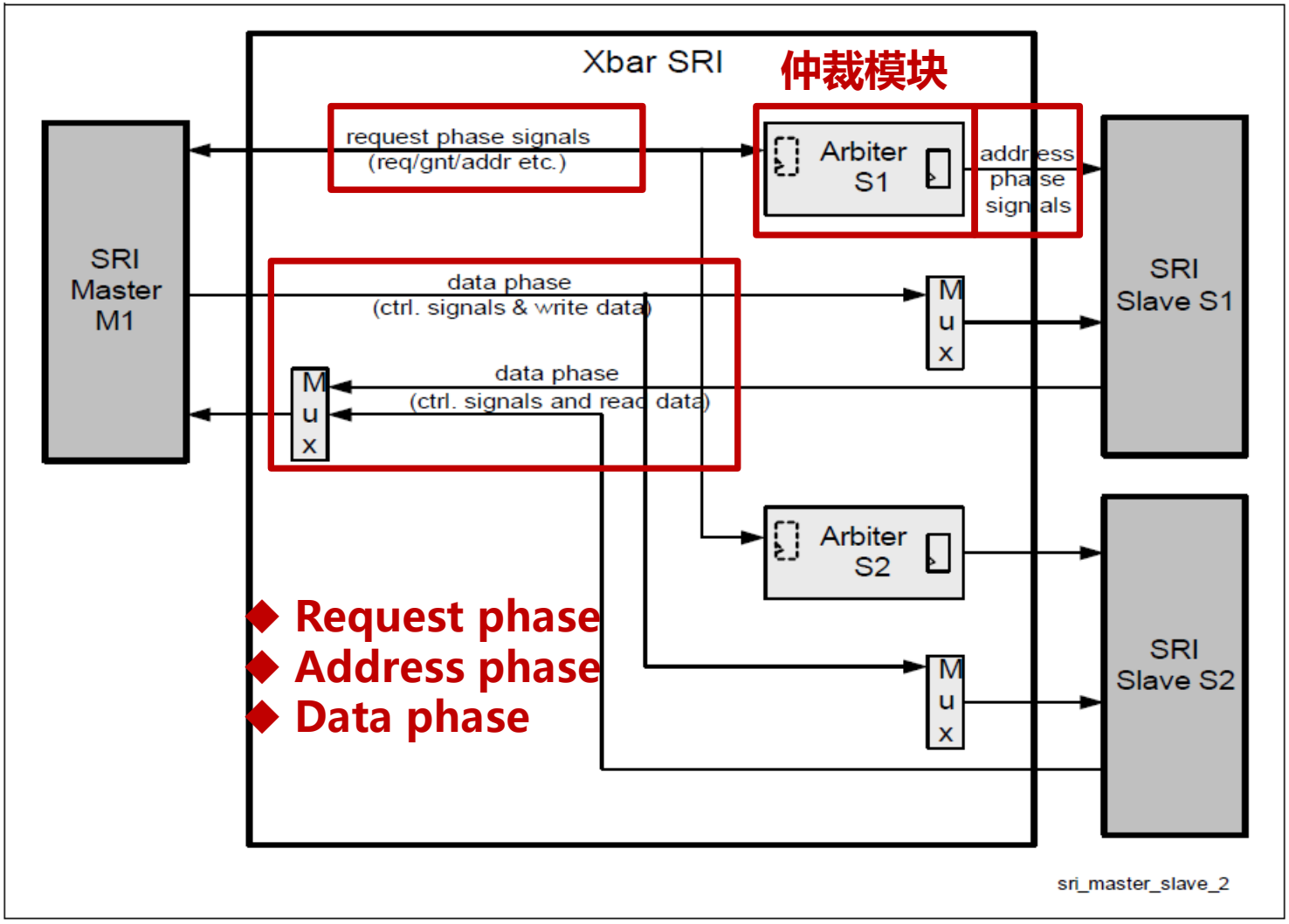


Figure 3-3 XBAR_SRI connections between SRI Master and SRI Slaves

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System Architecture of the TC21x/TC22x/TC23x

MCI: Master Connection Interface
SCI: Slave Connection Interface

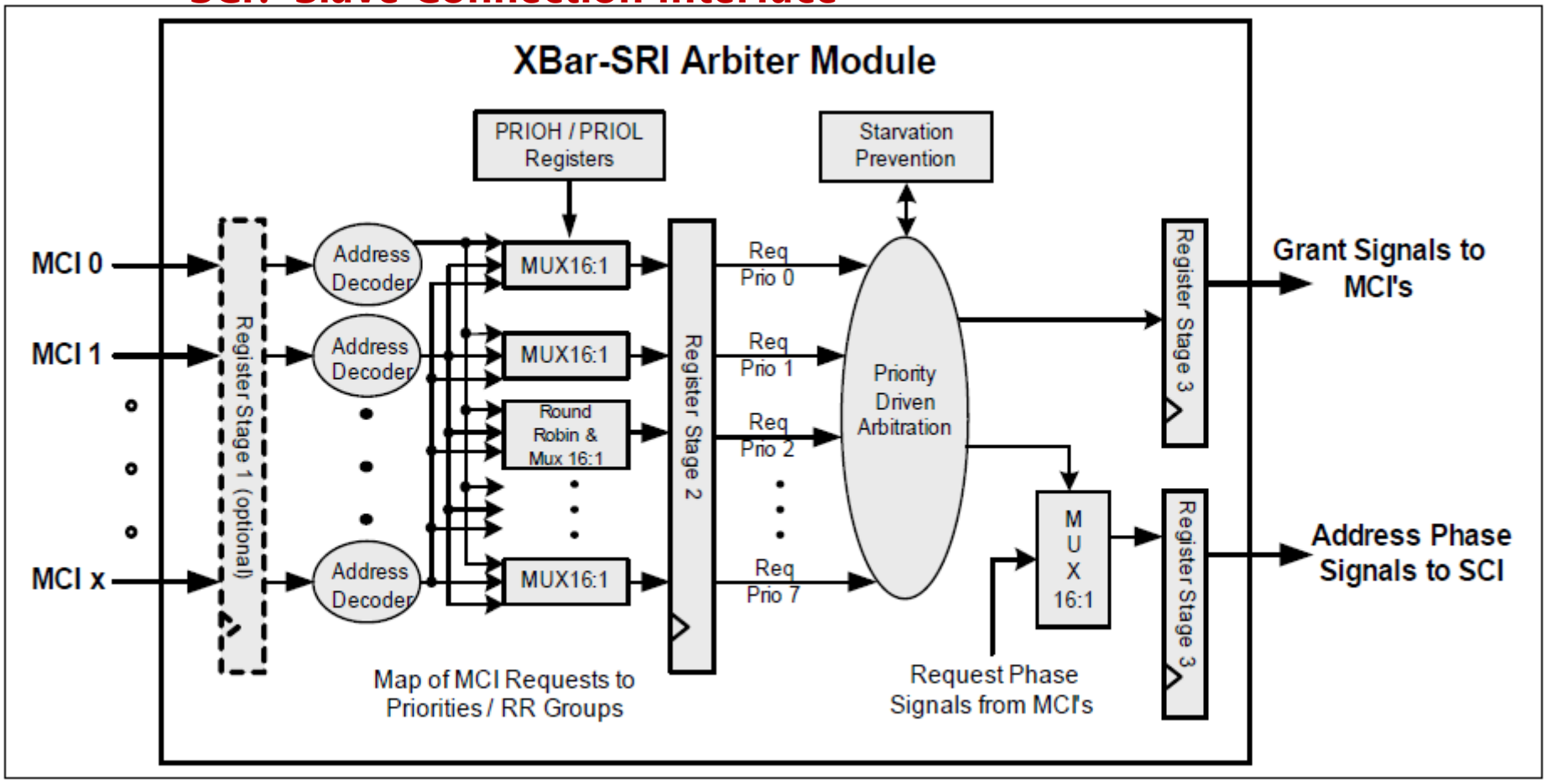


Figure 3-5 XBAR_SRI arbiter scheme

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Arbitration

The highest priority for an arbiter is 0 and the lowest is 7

Table 3-4 Default MCI Priority in the XBar_SRI Arbiters

Priority	Coding	Round Robin Group	Default Mapping:	Comments
0	000	-	-	
1	001	-	-	
2	010	Yes, max 8 master	MCI 0-7	Maximal 8 MCI can be mapped to the priority 2 e.g. can have the priority '010'.
3	011	-	-	
4	100	-	-	
5	101	Yes, max. 8 master	MCI 8-15	Maximal 8 MCI can be mapped to the priority 8 e.g. can have the priority '101'.
6	110	-	-	
7	111	-	-	

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Arbitration

The highest priority for an arbiter is 0 and the lowest is 7

Table 3-3 XBar_SRI Request Latency

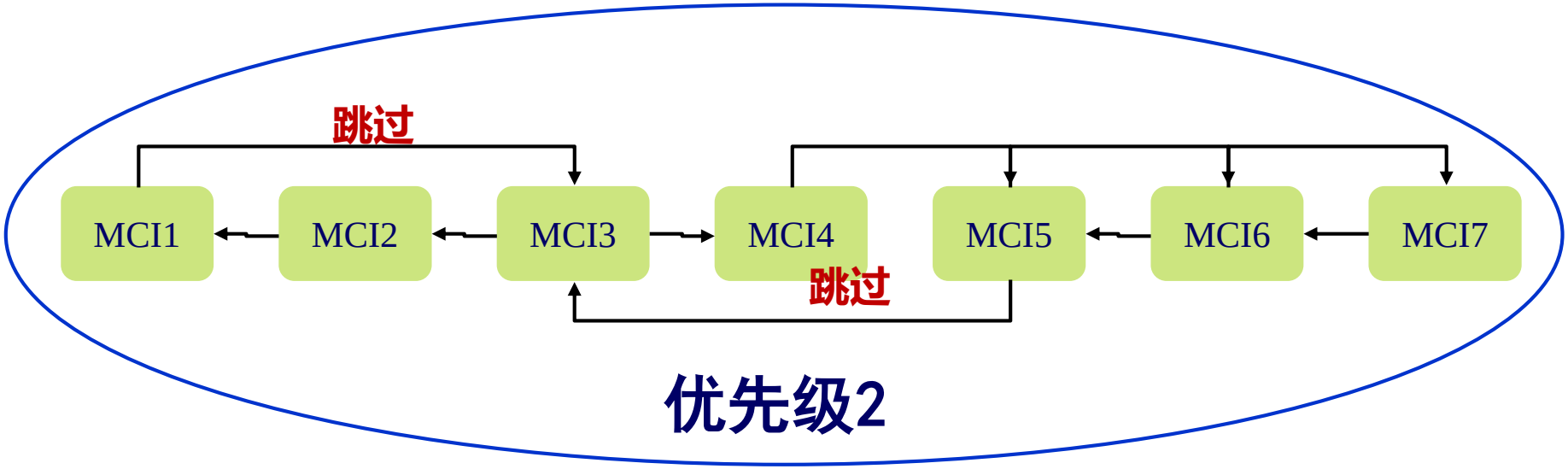
Clock Cycle Nr.:	Task(s)
0 (Default Slave)	sri_req_n is sampled at XBar_SRI MCI (Only valid for access to the XBar_SRI control registers)
1	address decoding, mapping of MCI's to priorities / round robin groups and starvation prevention. Round robin arbitrations, priority driven arbitration, mux request phase signals to the address phase registers masking sri_req_n from MCIx after granting the MCIx
2	sri_gnt_n to the requesting MCI, SRI address phase to SCI.

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Round Robin Group Arbitration：轮转法

优先级2和5的内部仲裁

After the winner of a round robin group is granted, the round robin group starts with a new arbitration round which means the requesting MCI's of the round robin group with the next highest MCI ID number will win the next round robin arbitration round. If there is no requesting MCI with a higher ID number in the round robin group, the algorithm will start with the MCI with the lowest MCI ID number that is mapped to the group, going to the next higher MCI ID number and so on.



确保总线不被优先机最高的MCI长期占据

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System Architecture of the TC21x/TC22x/TC23x

Table 3-6 Mapping of TC21x/TC22x/TC23x SRI master devices to MCI

XBar_SRI Master Connection Interface	MCI Number	Priority after Reset	Connected SRI master device
MCI 0	0	2	DMA
MCI 1	1	-	-
MCI 2	2	-	-
MCI 3	3	-	-
MCI 4	4	-	-
MCI 5	5	TC23x:2 TC22x: -	TC23x: SFI (HSM access to SRI) TC22x: -
MCI 6- MCI11	6-11	-	-
MCI 12	12	5	CPU0.DMI
MCI 13	13	5	CPU0.PMI
MCI 14- MCI 15	14-15	-	-

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System Architecture of the TC21x/TC22x/TC23x

Table 3-7 Mapping of TC21x/TC22x/TC23x SRI slave devices to SCI

XBar_SRI Slave Connection Interface (SCI)	Connected SRI master device
SCI 0	CPU0 (PSPR, DSPR, SFR, CSFR, PCache RAM, PTAG)
SCI 1-2	-
SCI 3	-
SCI 4	LMU (EMEM)
SCI 5	-
SCI 6	PMU0: DFlash, BootROM, CTRL Reg
SCI 7	PMU0: PFlash0
SCI 8- SCI 14	-
SCI 15	XBar_SRI Default Slave

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System Architecture of the TC21x/TC22x/TC23x

XBar_SRI Unit Register Overview

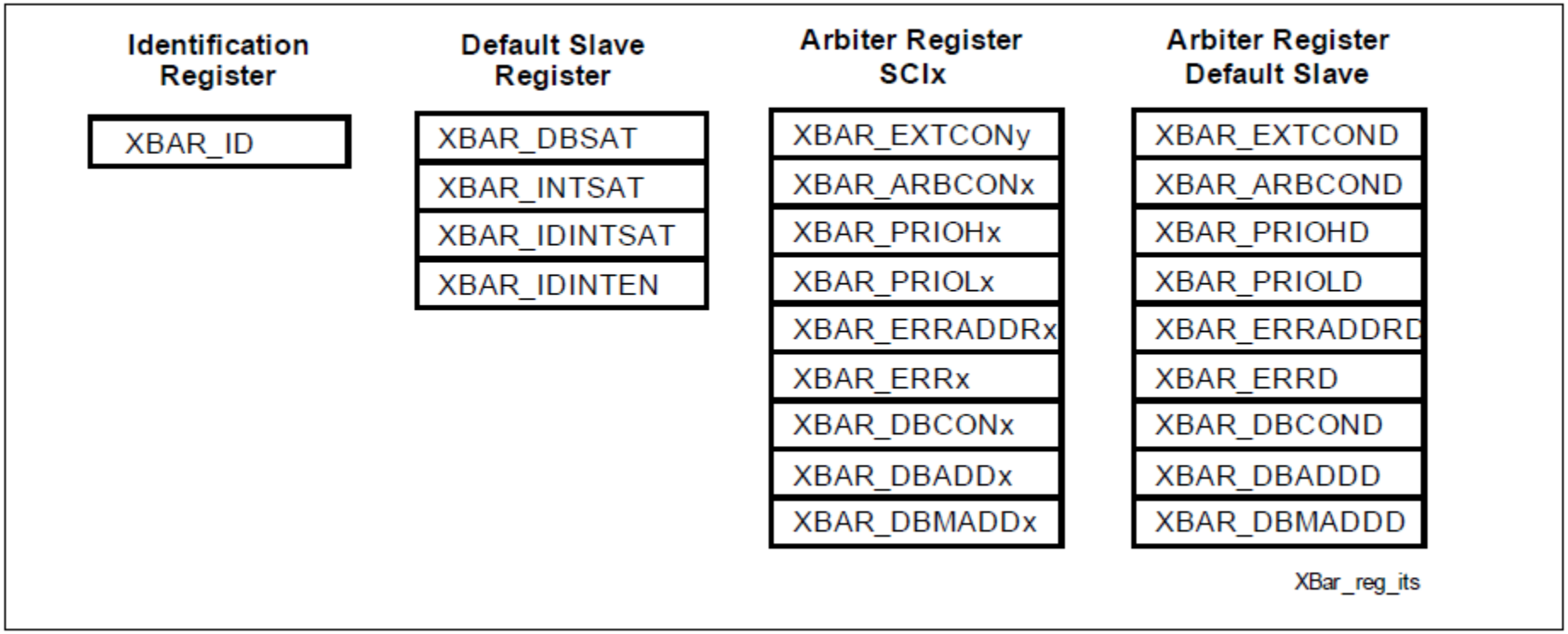


Figure 3-8 TC21x/TC22x/TC23x Control Registers

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System Architecture of the TC21x/TC22x/TC23x

List of Access Protection Abbreviations

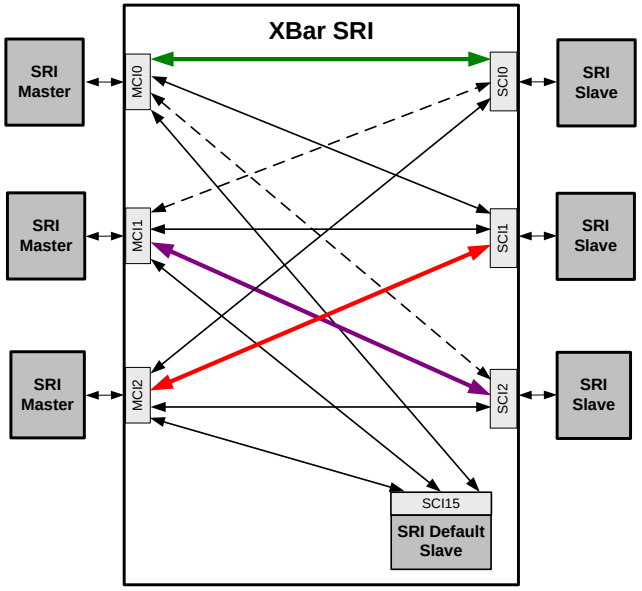
U	- User Mode
SV	- Supervisor Mode
BE	- Bus Error
nBE	- no Bus Error
P	- Access Protection, as defined by the ACCEN Register
E	- ENDINIT
SE	- Safety ENDINIT

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SRI Cross Bar (XBar SRI)

Shared Resource Interconnect (SRI) Cross Bar Interconnect:

- SRI protocol: 64 bit data bus, packet oriented protocol



Main SRI Cross Bar features:

- Point to point connections from masters to slaves
- Parallel / independent access
  - from different Master to different Slaves
- Parallel access to the same Slave
  - executed sequentially
  - configurable arbitration algorithms

8bit ECC covering 64bit data and bits [23..3] of address

- ECC protected (detection only):
  - Transaction address phases
  - Transferred read and write data

MASTER

SLAVE

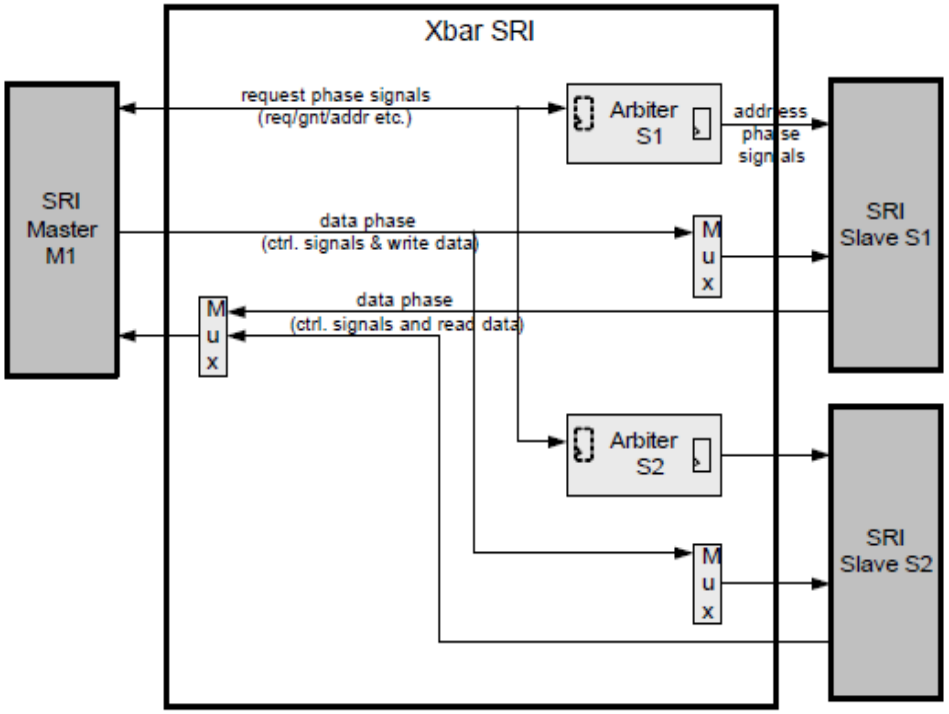
		DMA	HSSL	SFI	DAM	CPU1: DMI	CPU1: PMI	CPU2: DMI	CPU2: PMI	CPU0: DMI	CPU0: PMI
		MSC0	MSC4	MSC5	MSC6	MSC8	MSC9	MSC10	MSC11	MSC12	MSC13
CPU0	SSC0										
CPU1	SSC1										
CPU2	SSC2										
LMU	SSC4										
PMU0: Dflash /BROM	SSC6										
PMU0: Pflash0	SSC7										
PMU0: PFlash1	SSC8										

SFI: Bridge XBar_SRI <-> System Peripheral Bus
HSSL: High Speed Serial Link

SRI Cross Bar (XBar SRI)

Main SRI Cross Bar features (con't):

- Arbitration algorithm:
  - Priority driven (7-0)
  - Two Priorities (2, 5) can be used as Round Robin group
  - Each slave with own arbiter
  - Each slave with separate Master priority configuration
- Each slave interface with OCDS L1 debug capabilities
- MCDS: Any two slave interfaces can be observed in parallel



SRI Cross Bar (XBar SRI)

Cross Bar Interconnect Matrix shows:

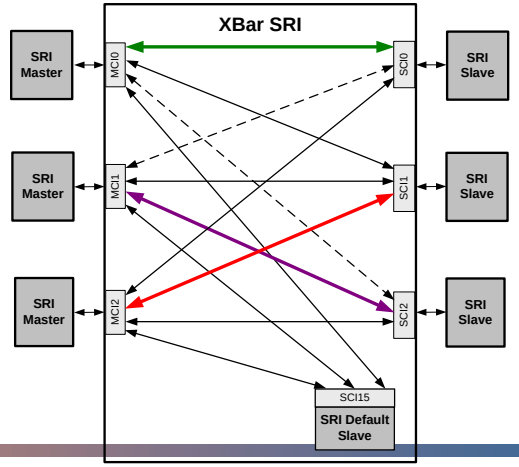
- Implemented Master / Slave interfaces
- Implemented Master / Slave connections

Example of parallel and independent access:

- CPU1 PMI is fetching code from PFlash0
- CPU2 PMI is fetching code from PFlash1
- CPU1 DMI is accessing CPU2 DSPR
- DMA / SDMA / HSSL with parallel access to different CPU/LMU SRAMs if available

TC27x: SRI Master / Slave Interconnection Matrix

		DMA	HSSL	SFI	DAM	CPU1: DMI	CPU1: PMI	CPU2: DMI	CPU2: PMI	CPU0: DMI	CPU0: PMI
		MSC0	MSC4	MSC5	MSC6	MSC8	MSC9	MSC10	MSC11	MSC12	MSC13
CPU0	SSC0										
CPU1	SSC1										
CPU2	SSC2										
LMU	SSC4										
PMU0: Dflash /BROM	SSC6										
PMU0: Pflash0	SSC7										
PMU0: PFlash1	SSC8										

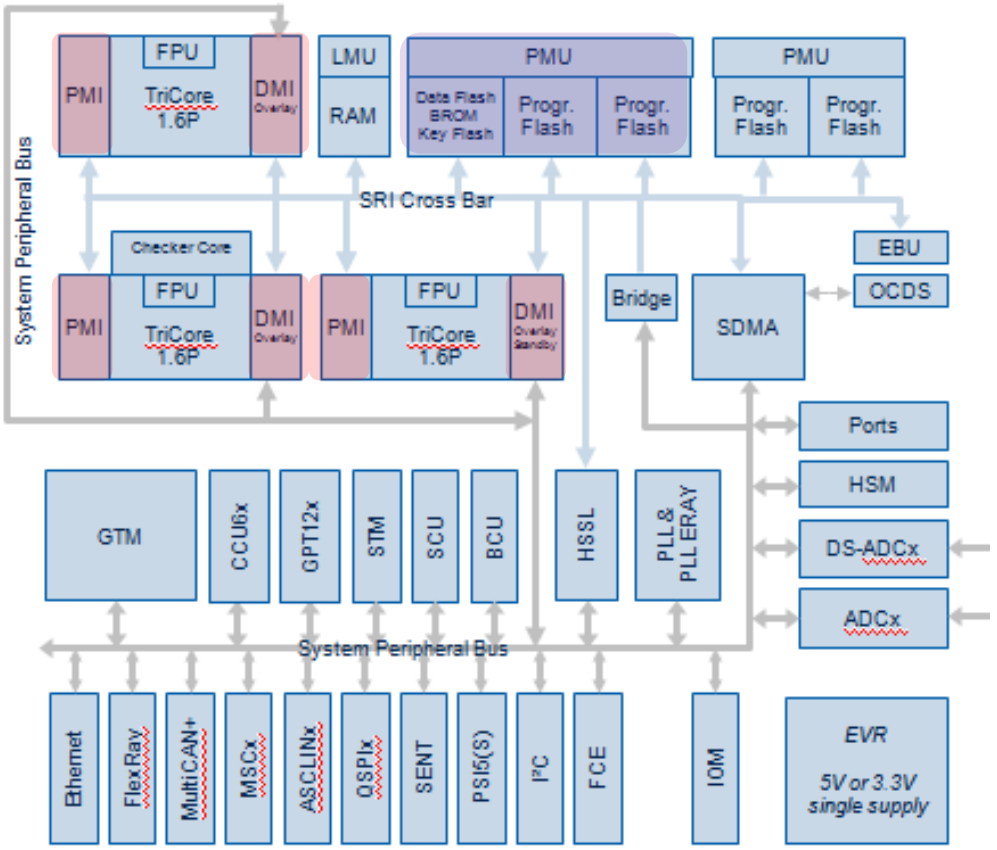


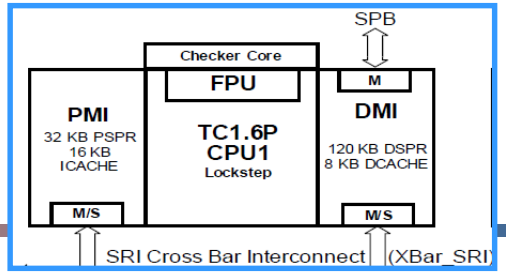
Architecture: Single / Multi-CPU

For the Aurix Multi-CPU architecture we have to cover additionally:

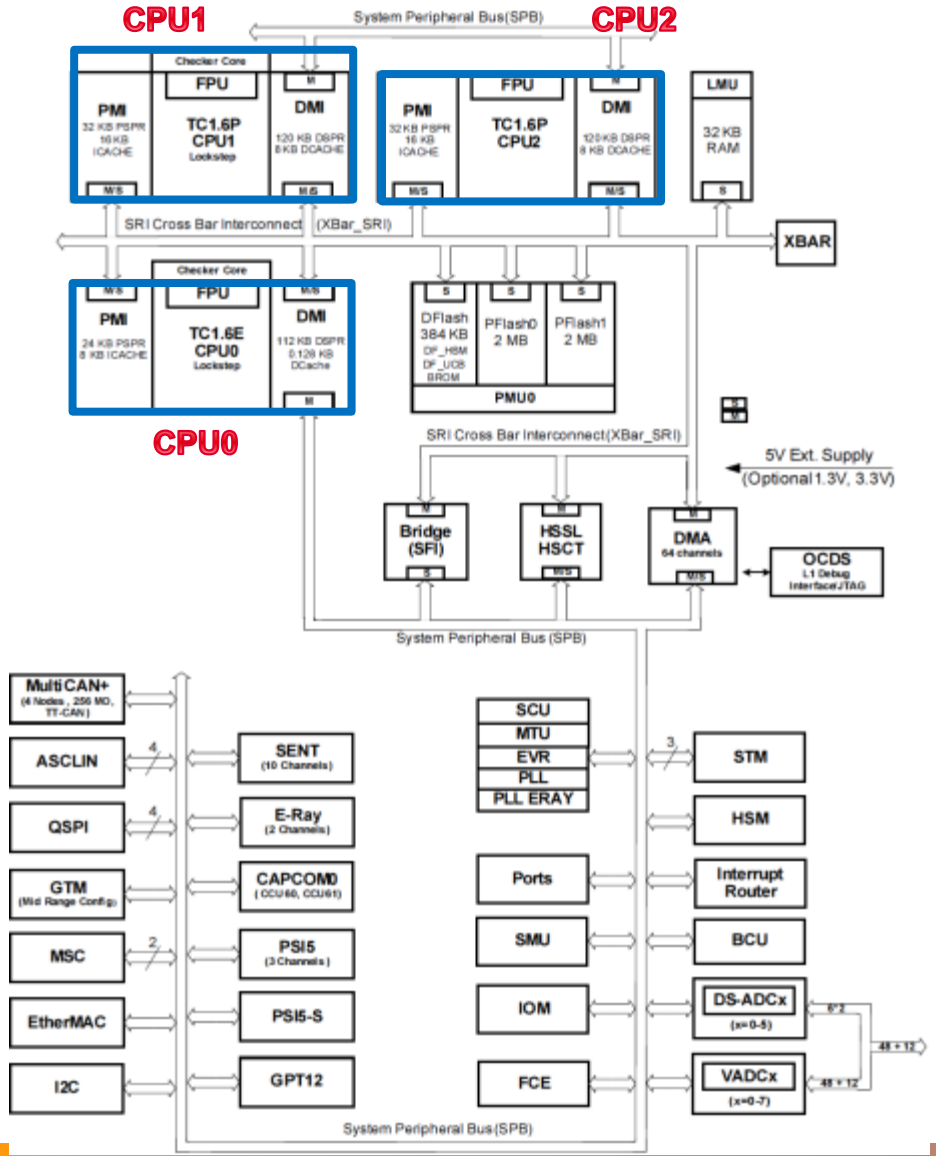
- Distributed Data Scratch Pad SRAMs (DSPRs)
  - CPUs data access to other SRAMs
- Distributed Program Scratch Pad SRAMs (PSPRs)
  - code fetch from other PSPRs
- CPUs executing Code from PFlash modules
  - More PFlash access conflicts
  - Reduced possibility for PMU to pre-fetch
- In parallel PFlash access from:
  - DMA to PFlash (e.g. periodic CRC checks)

- In addition:
  - CPU SW synchronization overhead
  - Extending single CPU to MultiCPU Interrupt System
  - Cross Bar Interconnect complexity





TC27x System Architecture

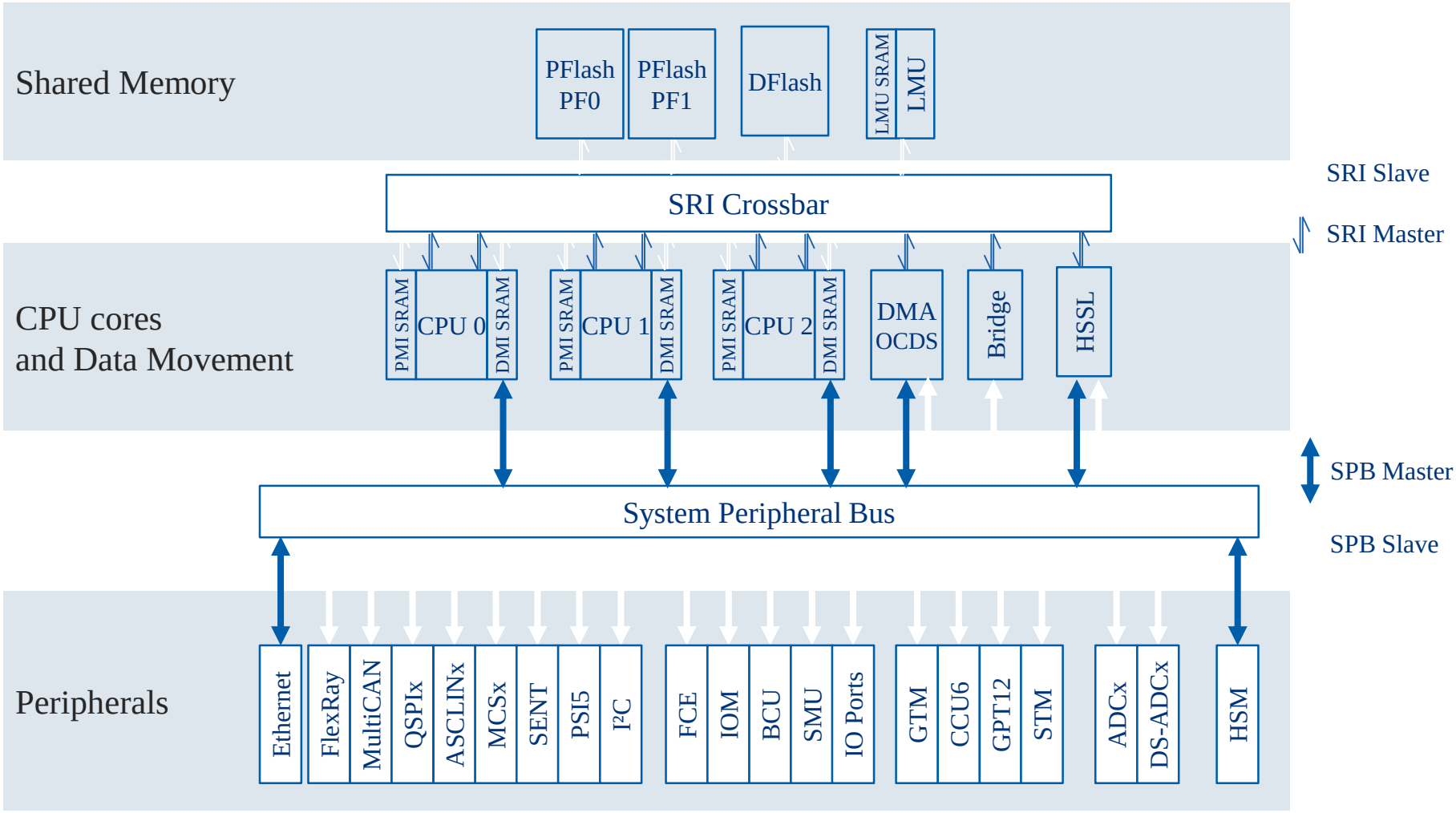


Addition in Multi-core architecture:

- Distributed **DSPRs** (In DMI)
  - CPUs with fast data access to other SRAMs
- Distributed **PSPRs** (In PMI)
  - High-perf. code fetch from other PSPRs
- CPUs executing Code from PFlash modules (PMU)
  - More PFlash access conflicts
  - Reduced possibility for PMU to pre-fetch
- In parallel PFlash access from:
  - DMA access to PFlash (e.g. periodic CRC checks)
- But also:
  - CPU SW synchronization overhead
  - Extending single core to MultiCore Interrupt System
  - Cross Bar Interconnect with 3 x TC1798 (AUDO MAX) complexity @300 MHz

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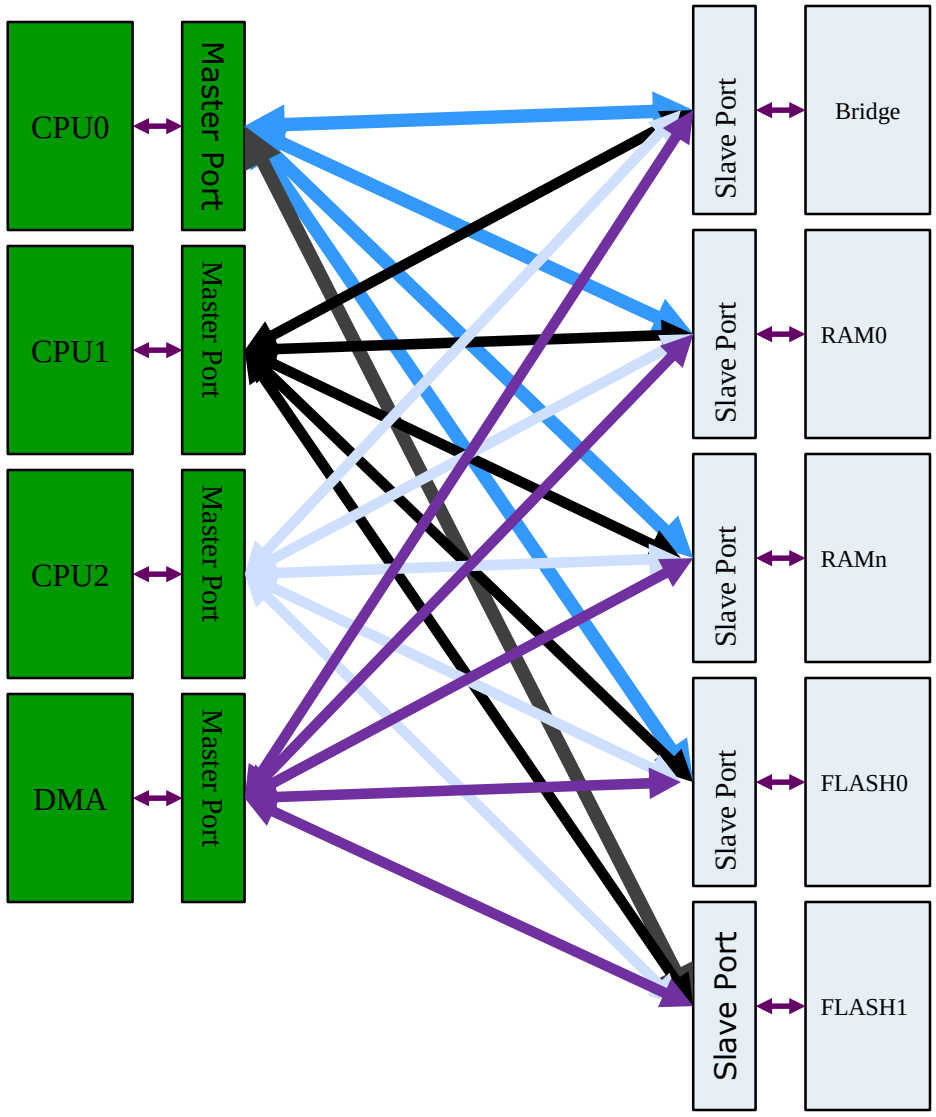
TC27x Bus Architecture



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SRI Crossbar for Multi-core

- One Master has direct access to several slave ports
- One slave serves one master at the same time
- Access Latency still can happen, if several masters using the same slave
- There is no access latency when different slaves are addressed
- Code and Data location has great influence on bus performance



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Multiple Parallel Access in SRI

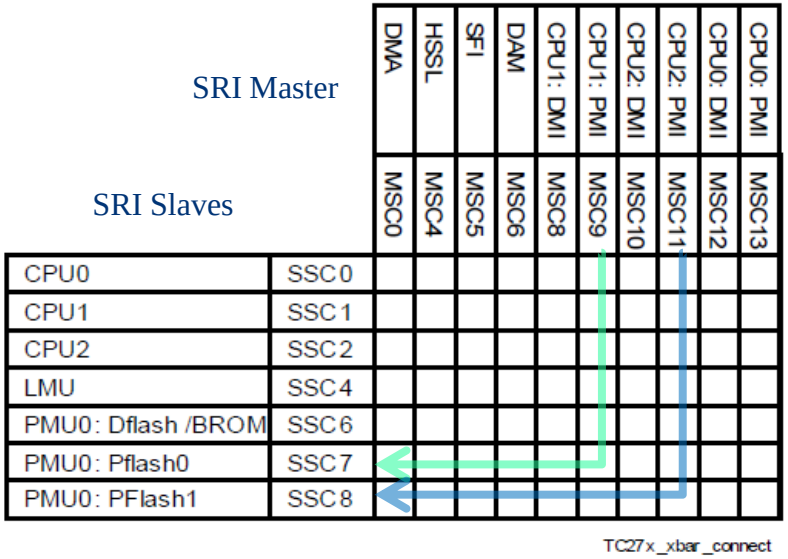
At The Same Cycle:

- CPU1 PMI is fetching code from PFlash0
- CPU2 PMI is fetching code from PFlash1
- CPU2 DMI is accessing the SPB
- DMA / SDMA / HSSL with parallel access to different CPU SRAMs

PMI: Program Memory Interface

DMI: Data Memory Interface

SFI: Bridge XBar_SRI <-> System Peripheral Bus (SPB)

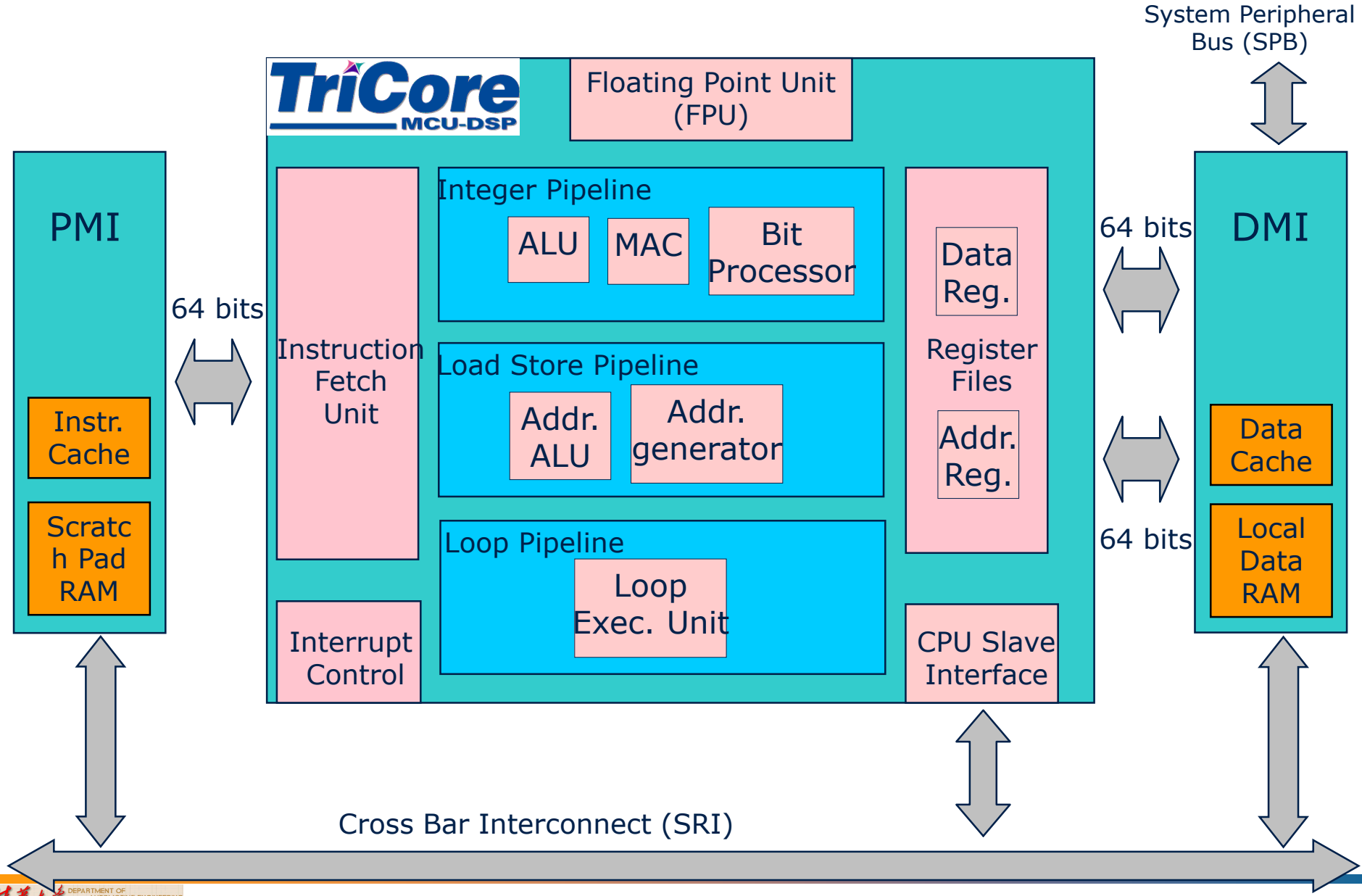


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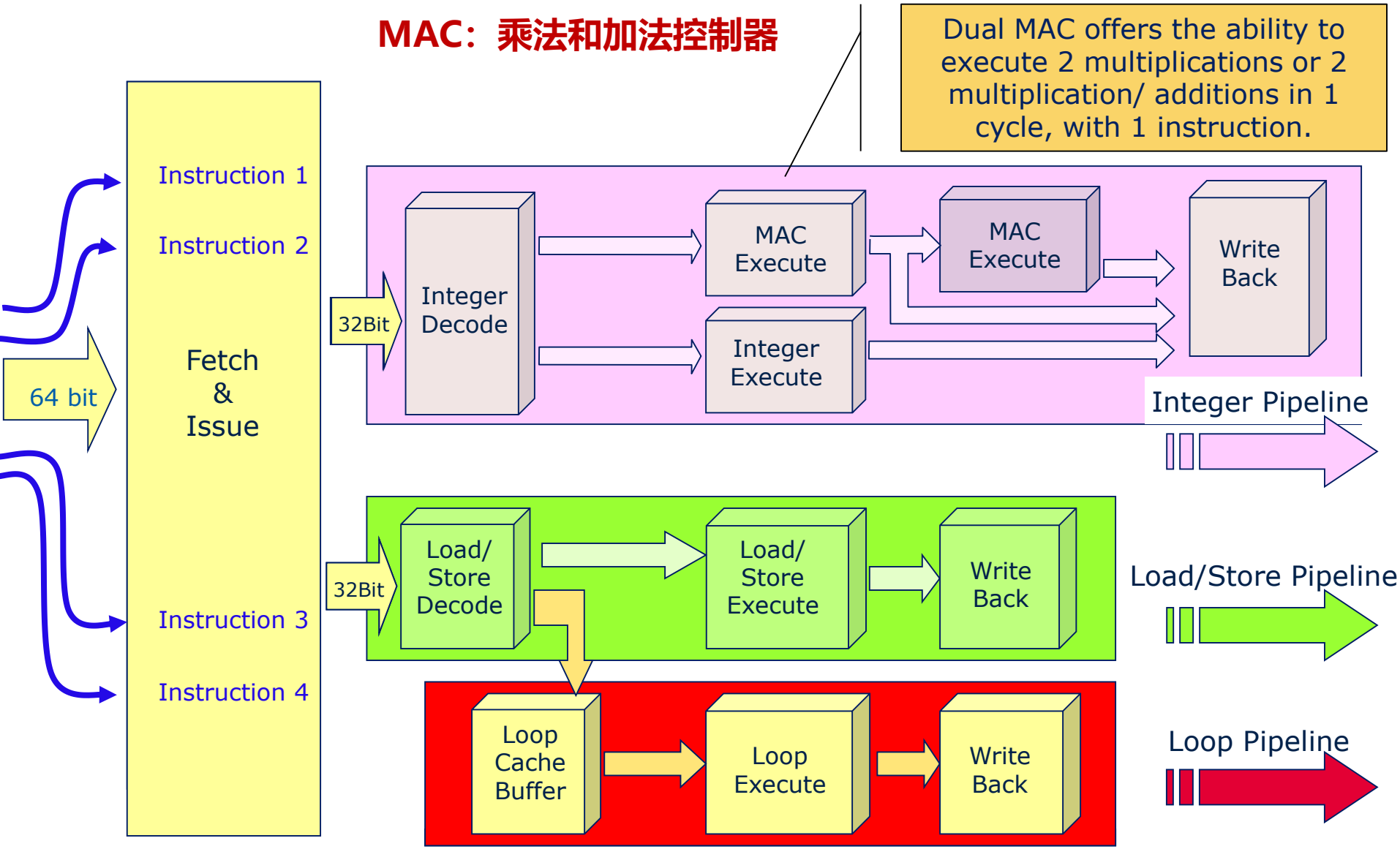
3 TC系列MCU的CPU与外围电路

- (1) MCU基本架构、CPU与外围模块访问模式
- (2) CPU架构、主要寄存器及工作原理
- (3) MCU内存管理

Core Subsystem



General Pipeline Overview
Four-Stage Super-Scalar triple issue TriCore™ Pipeline



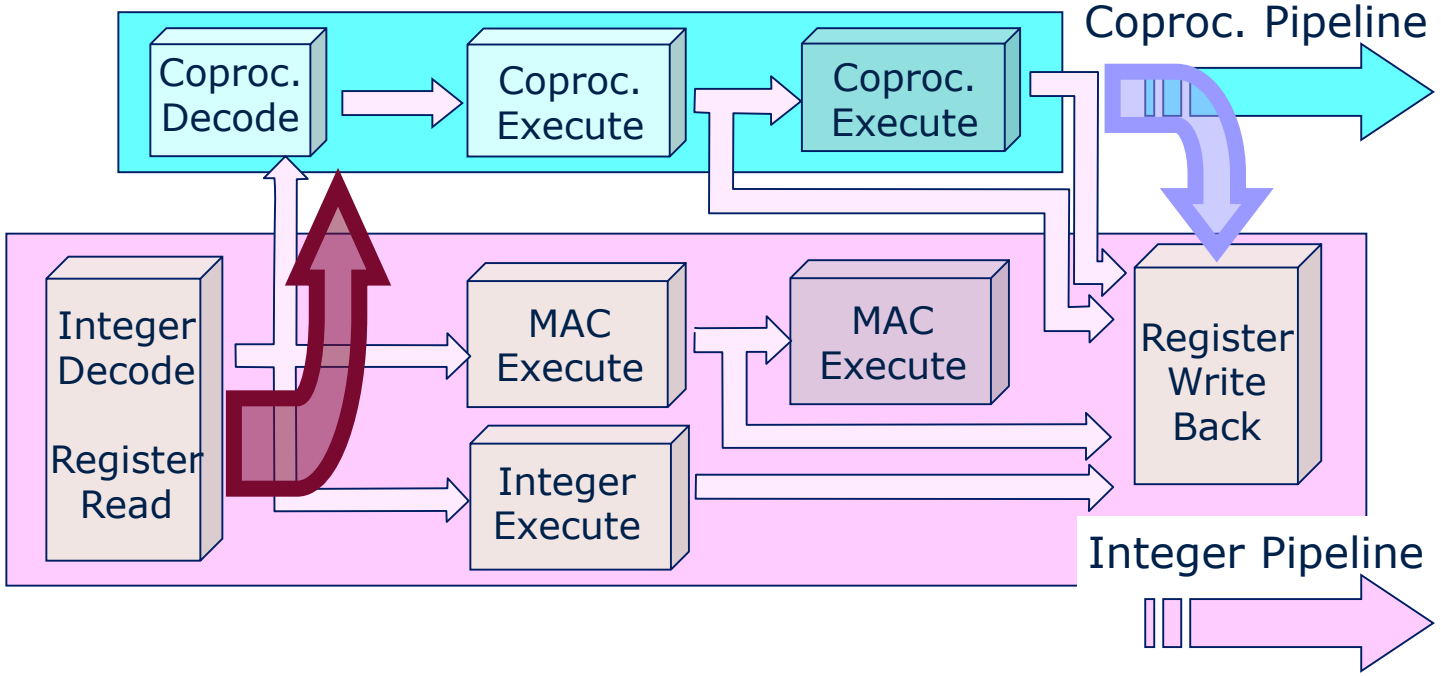
Pipeline - General

- **Integer pipeline:**
  - **Integer arithmetic and logic instructions.**
  - **Bit operations.**
  - **Divide and MAC instructions.**
  - **Etc.**

- **Load /Store pipeline:**
  - **Load / Store instructions.**
  - **Context operations.**
  - **Address arithmetic instructions.**
  - **Etc.**

- **Loop pipeline:**
  - **Loop instructions.**

General Floating Point Unit (FPU) Overview With Four Stage Pipeline



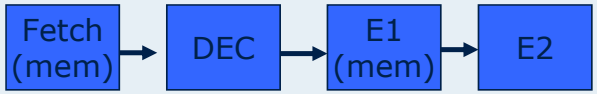
Floating Point Instructions	
ADDF	2
CMP.F	1
DIV.F	15
FTOI	2
FTOIZ	2
FTOQ31	2
FTOQ31Z	2
FTOU	2
FTOUZ	2
ITOF	2
MADD.F	3
MSUB.F	3
MUL.F	2
Q31TOF	2
QSEED.F	1
SUB.F	2
UPDFL	–
UTOF	2

Passing Instructions and Register Content to Coprocessor

Passing result from Coprocessor to Register

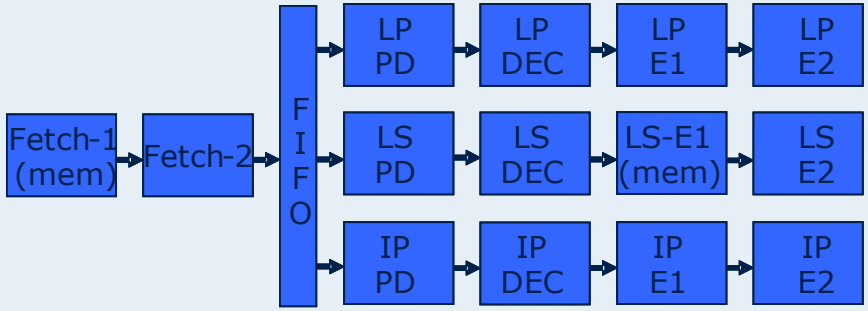
Infineon's TriCore Processor for AURIX Efficiency and Performance

TriCore 1.6Efficiency



- High **efficiency / low power** architecture
  - **Scalar** Harvard
  - **4 pipeline stages** for up to 200MHz
- Identical instruction set as TC1.6P
- **1.4 DMIPS/MHz**
- Instruction cache

TriCore 1.6Performance



- High **performance** architecture
  - **Superscalar** Harvard
  - **6 pipeline stages** for up to 300MHz
- Identical instruction set as TC1.6E
- **1.6 DMIPS/MHz**
- Instruction and data cache



32bit Floating Point Unit in all CPUs:

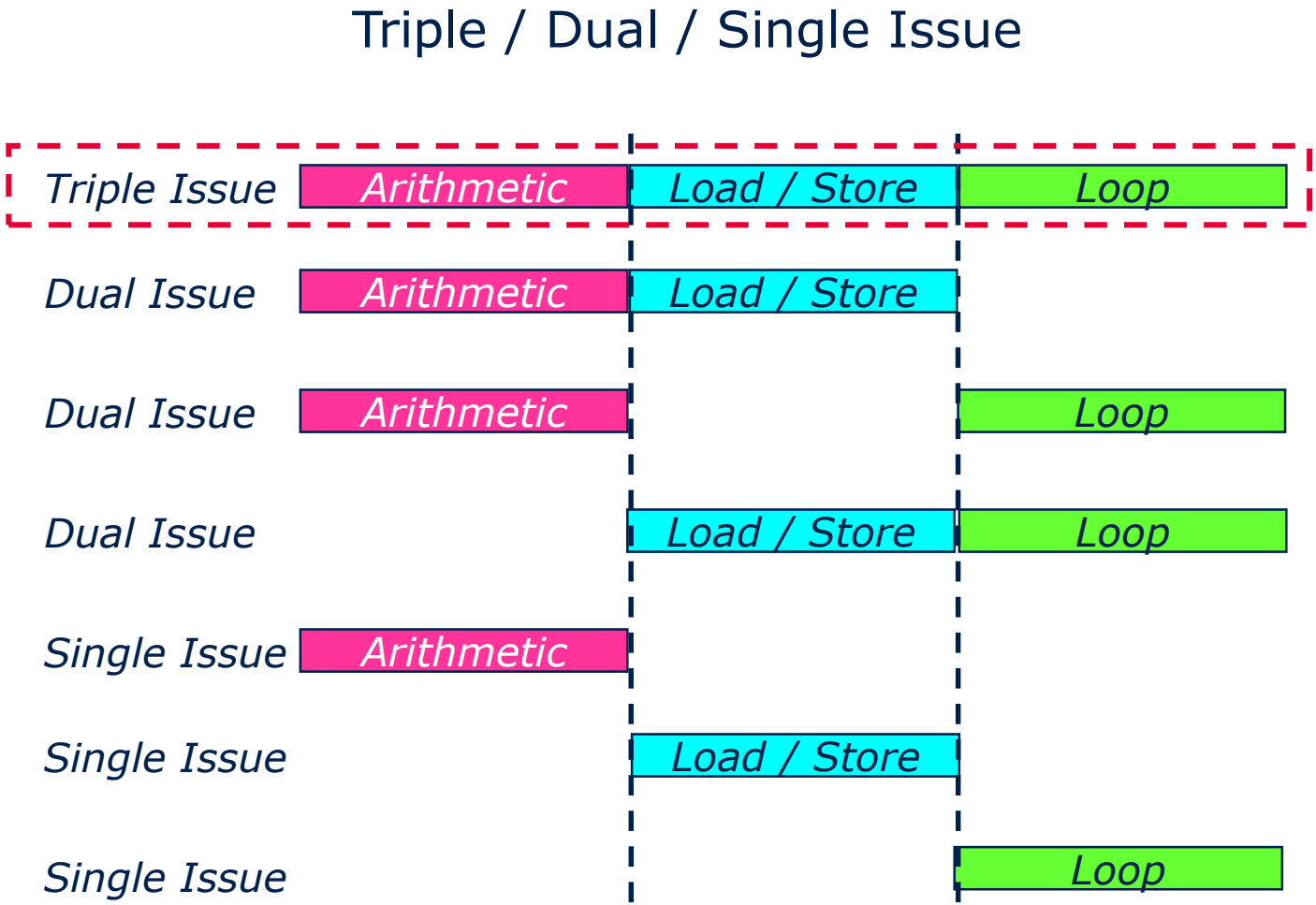
- Single precision according to IEEE-754
- 2 FLOPs per cycle (pipelined)

基本时钟频率（对应一条指令）：200/300MHz

节拍频率：200*4/300*6MHz

Super-scalar execution

~1.3 instr/cycle



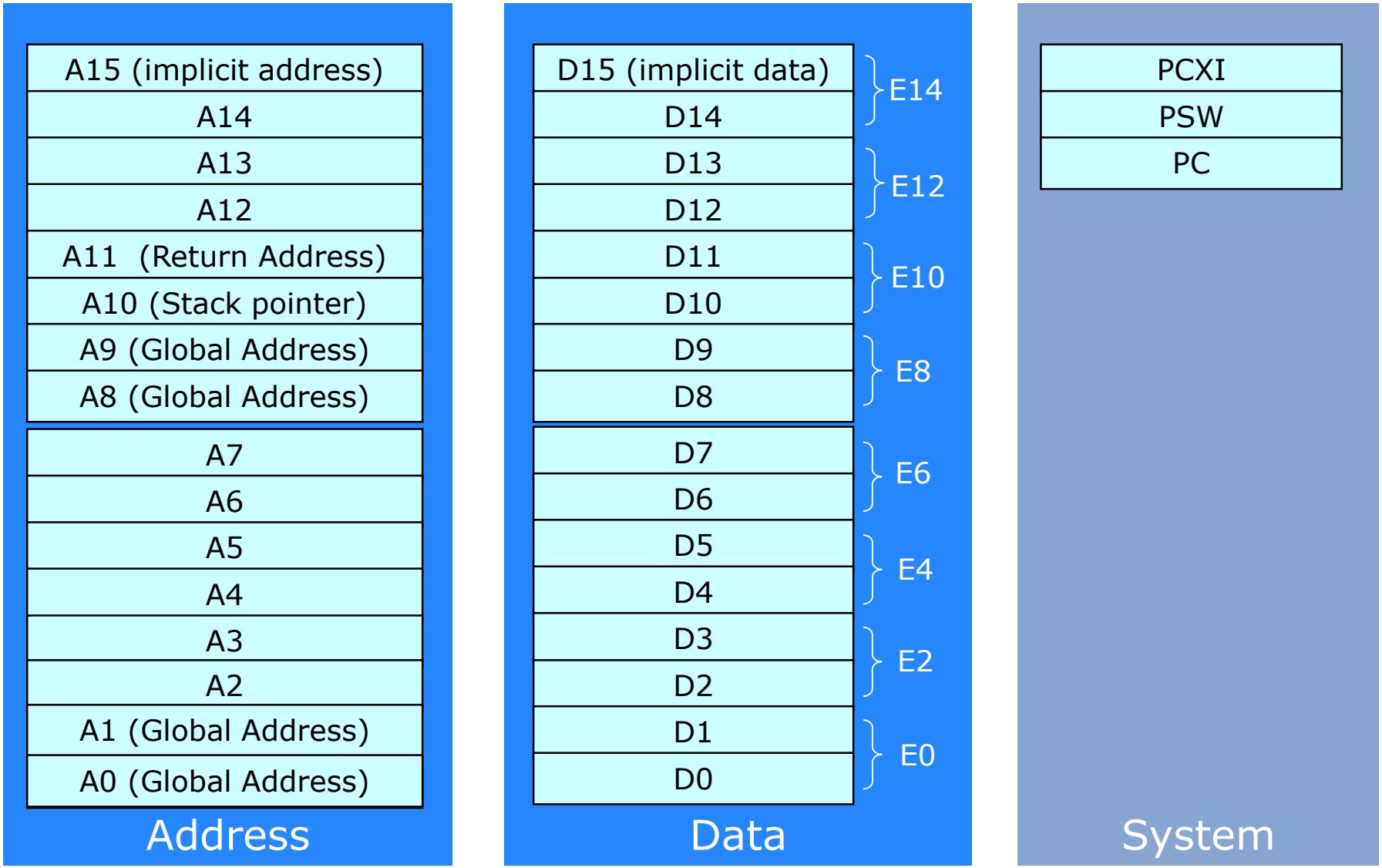
3 instructions in a single cycle is possible provided the instructions follow this sequence
1) Arithmetic
2) Load/Store
3) Loop

TriCore Overview

PCP

	TC 1.3	TC 1.3.1	TC 1.6	TC 1.6Performance (version TC 1.6.1)	TC 1.6Efficiency (version TC 1.6 Scalar)
Architecture	32-bit superscalar Harvard	32-bit superscalar Harvard	32-bit superscalar Harvard	32-bit superscalar Harvard	32-bit scalar Harvard
Max. Frequency	150 MHz	180 MHz	300 MHz	300 MHz	200 MHz
Pipeline stages	4	4	6	6	4
DMIPS / MHz	1,55	1,68	1,7	1,7 – 2,3	1.4 – 1,7
Instruction cache	Two way set associative, 256-bit cache line length	Two way set associative, 256-bit cache line length	Four way set associative, 256-bit cache line length	Two way set associative, 256-bit cache line length	Two way set associative, 256-bit cache line length
Data cache	Not in Automotive Products	Two way set associative 128-bit cache line length	Four way set associative 128-bit cache line length	Two way set associative 256-bit cache line length	4 line read buffer 256-bit cache line length
HW int. division unit	No	No	Yes	Yes	Yes
MAC units	dual 16x16 multipliers	dual 16x32 multipliers	dual 16x32 multipliers	dual 16x32 multipliers	dual 16x32 multipliers
FPU	Single precision (32-bit)	Single precision (32-bit)	Single precision (32-bit)	Single precision (32-bit)	Single precision (32-bit)
FPU performance	0,66 FLOPs/cycle	0,66 FLOPs/cycle	2 FLOPs/cycle (pipelined)	2 FLOPs/cycle (pipelined)	2 FLOPs/cycle (pipelined)
User modes	Supervisor, user 1, user 0	Supervisor, user 1, user 0	Supervisor, user 1, user 0	Supervisor, user 1, user 0	Supervisor, user 1, user 0
Memory protection	2 protection sets (overlaid HW breakpoints)	4 protection sets (overlaid HW breakpoints)	4 protection sets (separate HW breakpoints)	4 protection sets (separate HW breakpoints)	4 protection sets (separate HW breakpoints)
Instruction set	16/32 bit instructions	16/32 bit instructions	16/32 bit instructions	16/32 bit instructions	16/32 bit instructions
Compatibility	TC1.2 compatible	TC1.3 compatible	TC1.3.1 compatible	TC1.6 compatible	TC1.6 compatible
Other features	■ static branch predictor	■ dynamic branch predictor	■ Enhanced branch pre-diction (branch history and target buffers)	■ Dual core support ■ Safety support	■ Dual core capable ■ Power consumption optimized ■ Safety support
Status	Production (Audo-NG)	Production (AudoFuture)	Development for 90nm	Development for 65nm	Development for 65nm

Architectural Registers



Core Registers

General Purpose Registers	
D0 – D15	Data Registers.
A0 – A15	Address Registers.

System Registers	
PC	Program Counter
PSW	Program Status Word
SYSCON	System Control Registers
PCXI	Previous Context Information.

Context Management	
FCX	Free CSA List Head Pointer
LCX	Free CSA List Limit Pointer

CPU Interrupt and Trap Control	
ICR	Interrupt Control Reg.
BIV	Base Address of Interrupt Vect. Table.
BTV	Base Address of Trap Vect. Table.

Memory Protection	
DPRx_0.3	Data Segment Protection Regs.
DPMx	Data Protection Mode Reg. x
CPRx_0..1	Code Segment Protection Regs.
CPMx	Code Protection Mode Reg. x

Stack management	
ISP	Interrupt Stack Pointer

Core Special Function Registers (CSFRs) can only be accessed by special read/write instructions:
→ MTCR: move to core special function register (write access, supervisor mode only)
→ MFCR: move from core special function register (read access)

PSW register

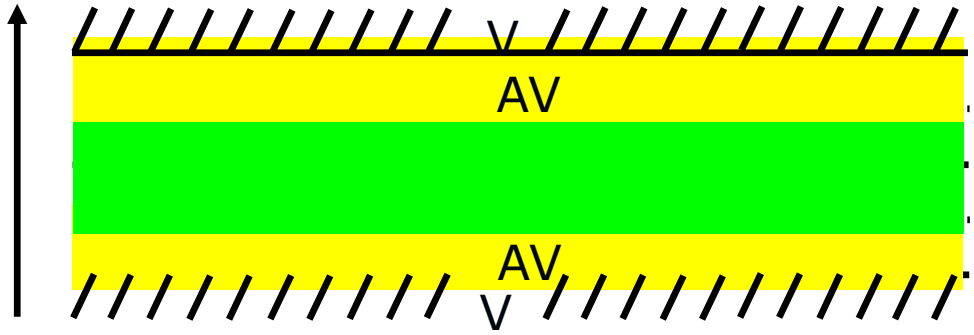
C: Carry
V: Overflow
SV: Sticky Overflow
(Flag is only clear by special Instruction)
AV: Advanced overflow
SAV: Sticky Advanced Overflow
(Flag is only clear by special Instruction)

PSW
Program Status Word **Reset Value: 0000 0B80$_H$**

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
C	V	SV	AV	SAV	0										
rwh	rwh	rwh	rwh	rwh	r										

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0		PRS		IO		IS	GW	CDE	CDC						
r		rwh		rwh		rwh	rwh	rwh	rwh						

CDC: Call Depth Counter
(with call depth limit of 63)
CDE: CDC enable



Overflow and Advanced Overflow

GW: Global Address write permission

IS: Interrupt stack control
(using Interrupt stack or user stack when a enter a ISR)

IO: User mode
(User0, User1 or Supervisor mode)

PRS: Protection register set
(to select the memory protection set)

What is a context?

- In an embedded real time system, independent tasks switching occur frequently (in response to an interrupt, RTOS management, etc).
- The task's context represents the state of this task.
- Basically, the context is everything the μC needs to know in order to start (or re-start) a task.

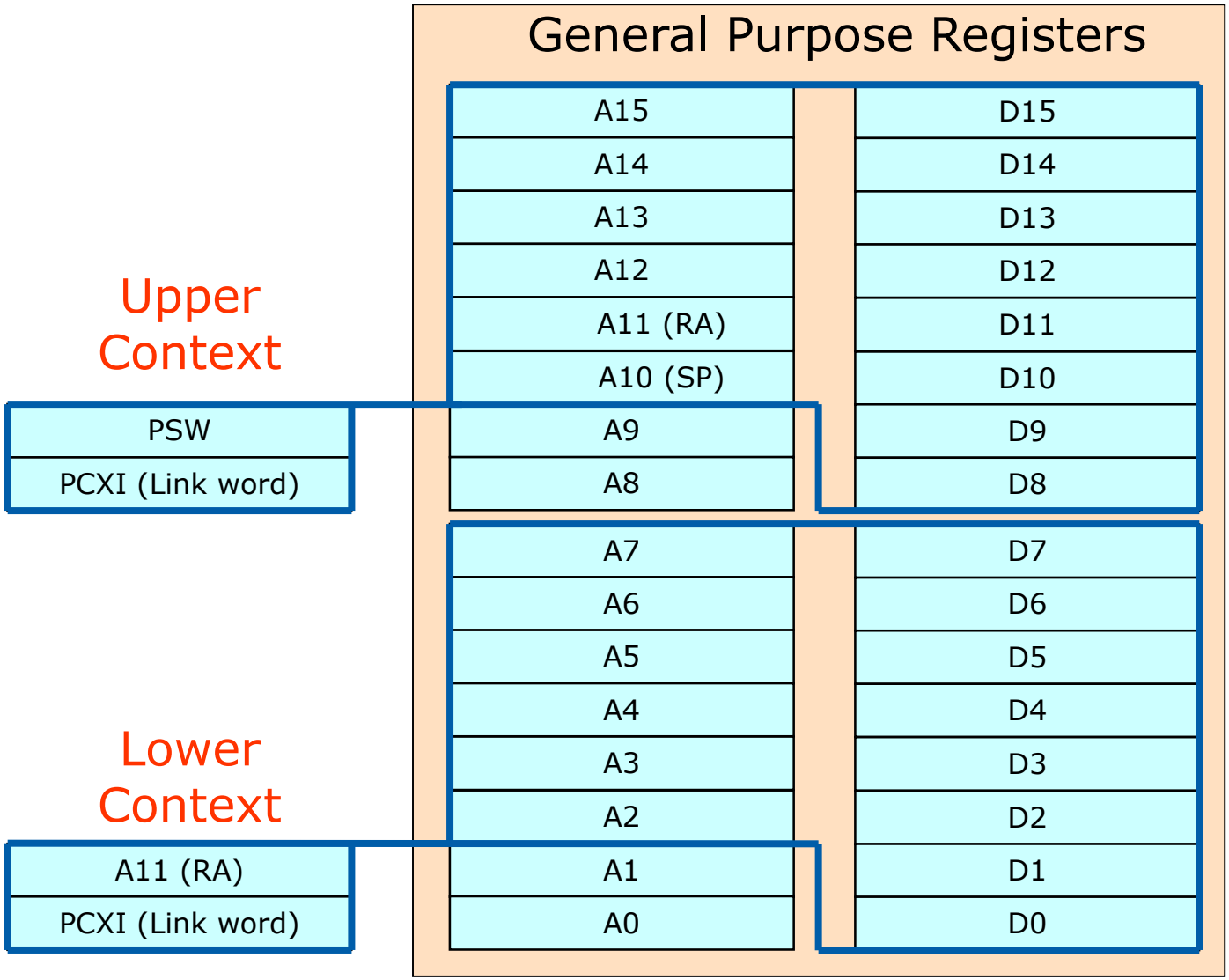
Upper and Lower context

- TriCore defines two contexts:
  - The upper context (task specific)
  - The lower context (for parameter passing).

- The upper context is automatically saved on call, interrupt or trap.

- The lower context has to be saved explicitly with an instruction.

Upper and Lower Contexts



Context Saving and restoring

Save

Restore

Event / Instruction	Saved context	Event / Instruction	Restored Context
Interrupt	Upper	RFE	Upper
Trap	Upper	RFE	Upper
CALL	Upper	RET	Upper
BISR	Lower	RSLCX	Lower
SVLCX	Lower	RSLCX	Lower
STLCX	Lower	LDLCX	Lower
STUCX	Upper	LDUCX	Upper

Automatically saved / restored

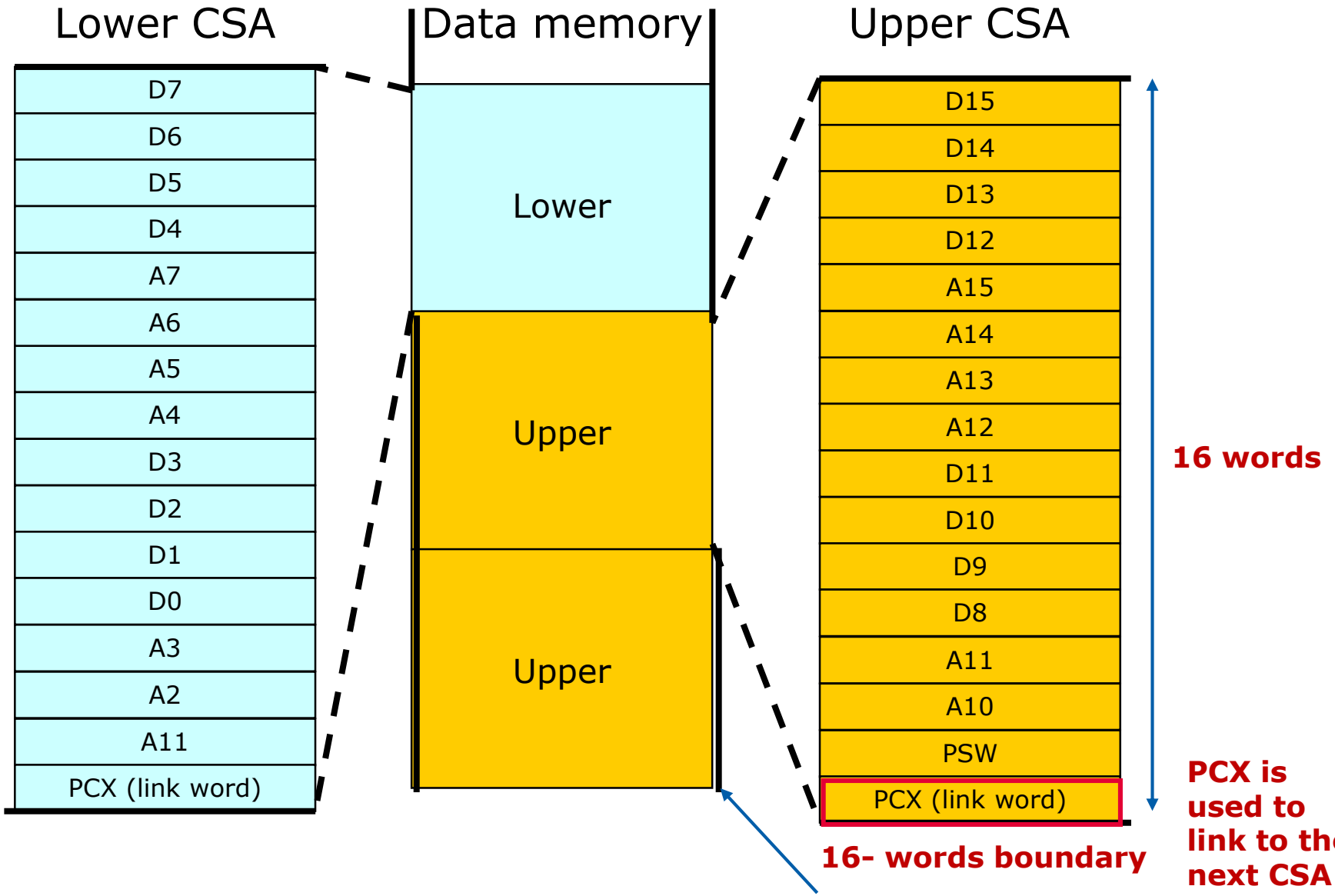
Saving to CSA

Saving to Absolute Address

Context Save Areas

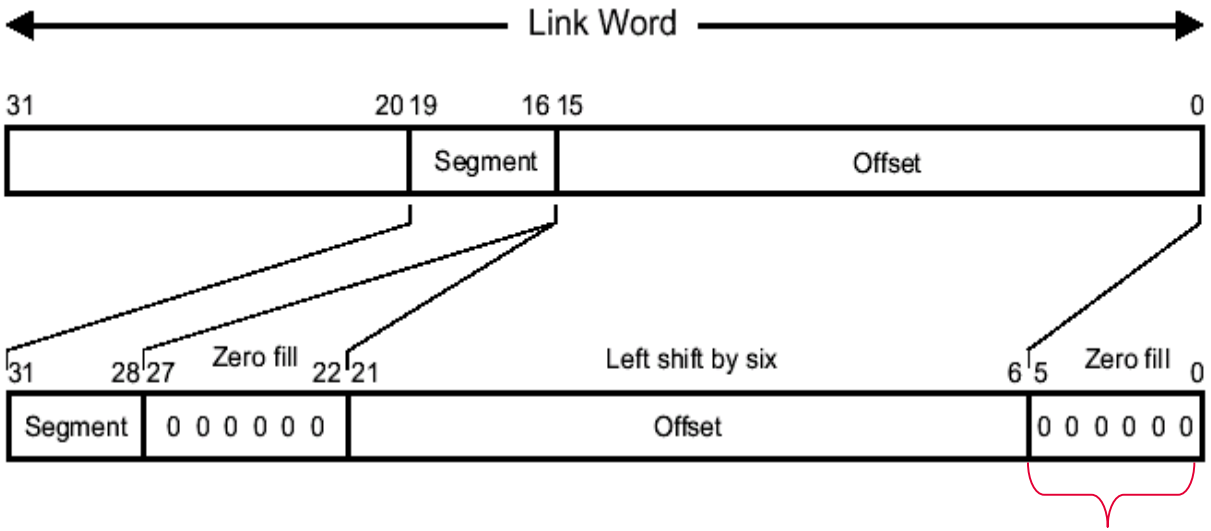
- Context Save areas can hold 1 upper or 1 lower context.
- CSA are aligned on a 16-word boundary.
- Trap generated when context list becomes depleted

Context Save Areas



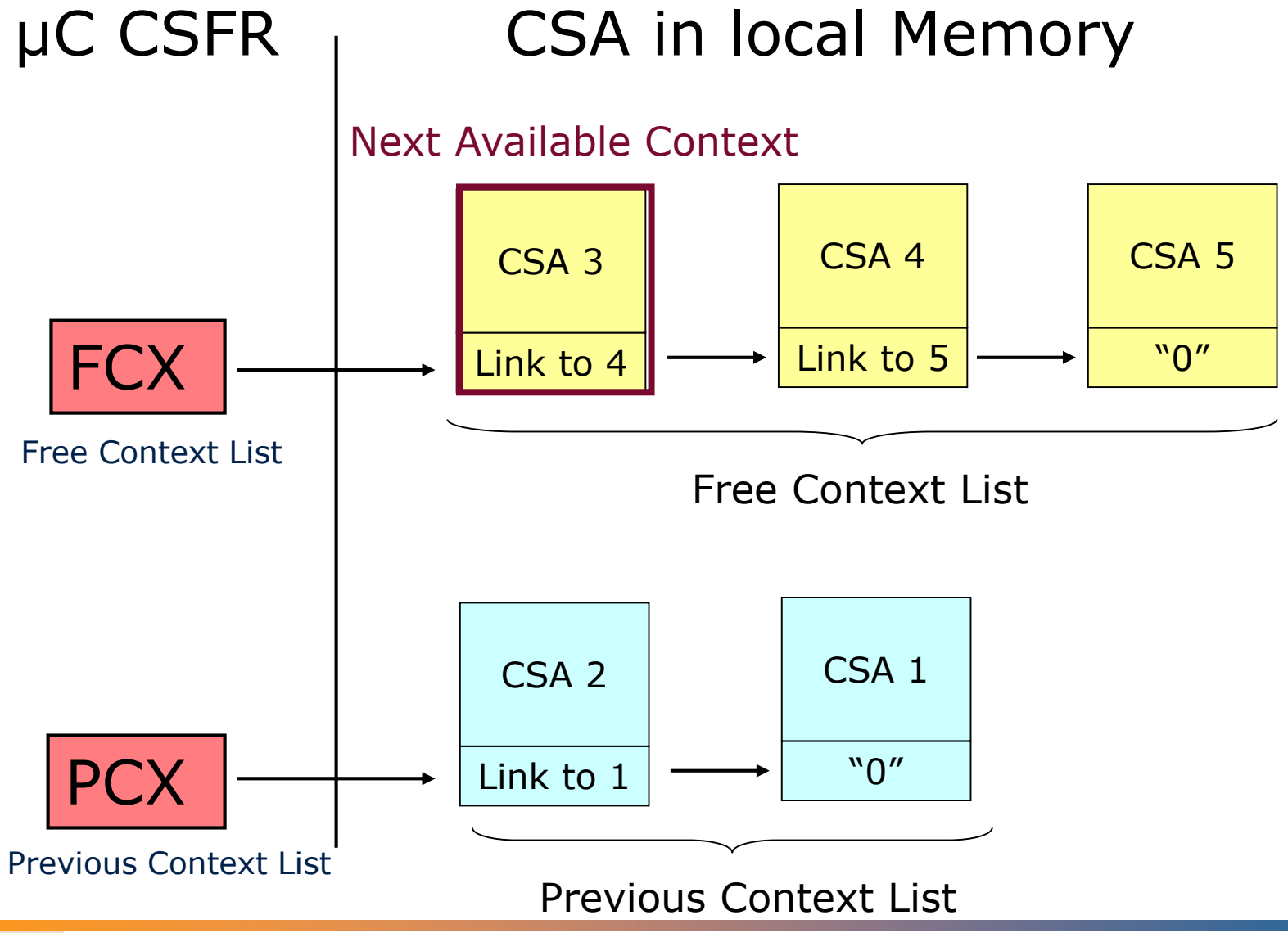
Context Save Areas

- What is PCX Link word?

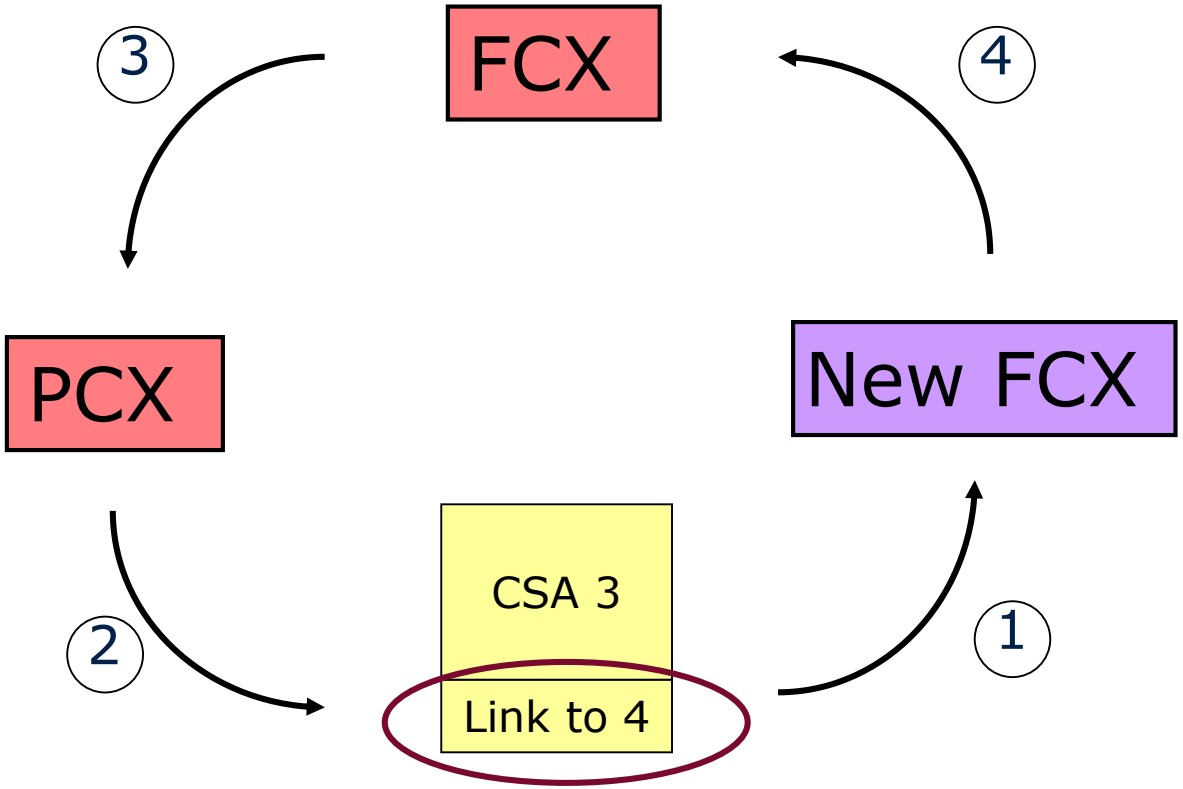


- CSAs are linked via their link word in lists:
  - "Free Context List" for unused CSA. FCX points to the first available CSA
  - "Previous Context List" for used CSA. PCX points to the most recently saved CSA.
- Hardware automatically handles allocation & freeing of CSAs.

CSAs prior to Context Save



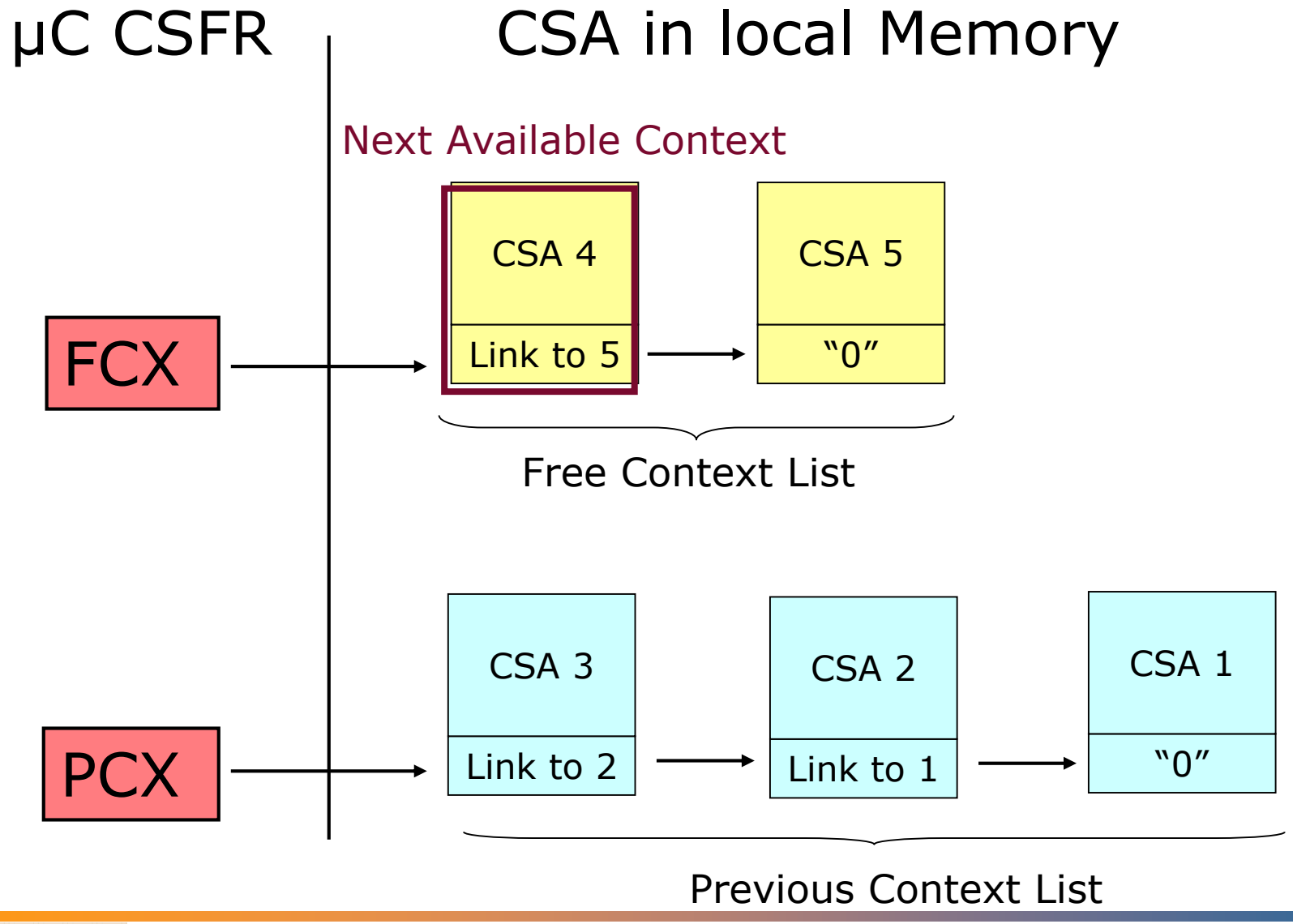
Context Save Processing



When an interrupt / call / trap happened,

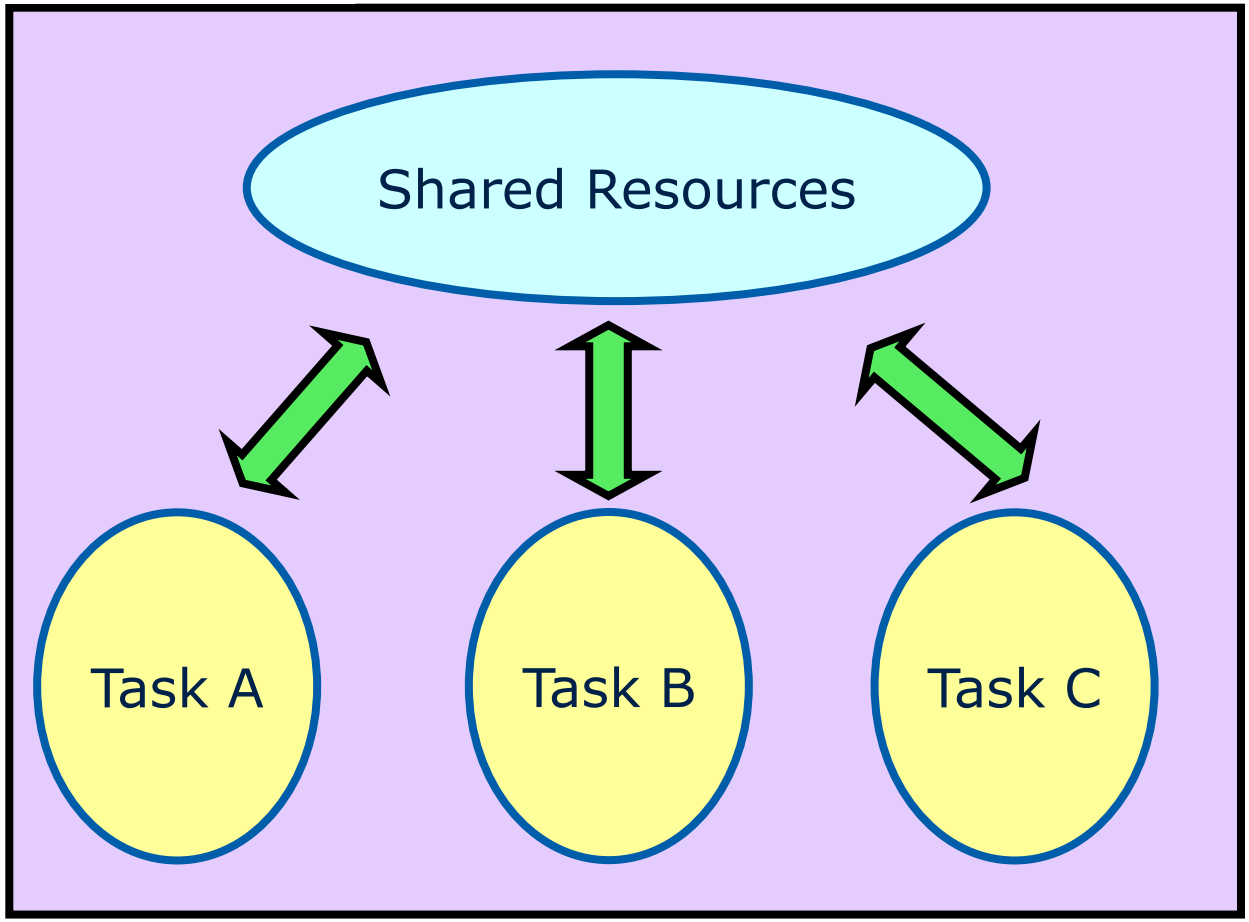
1) New FCX = Link word in CSA3 = CSA4
2) PCXI of CSA3 = PCX (pointing to CSA2)
3) PCX = FCX (pointing to CSA3)
4) FCX = New FCX (pointing to CSA4)

CSAs After Context Save



Multitasking system

TriCore MCU-DSP



Protection System

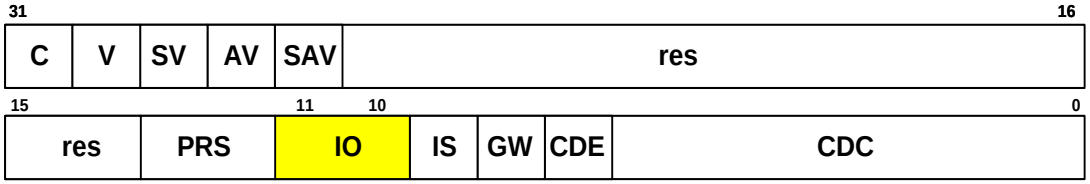
- Embedded applications constantly increase in size and complexity.

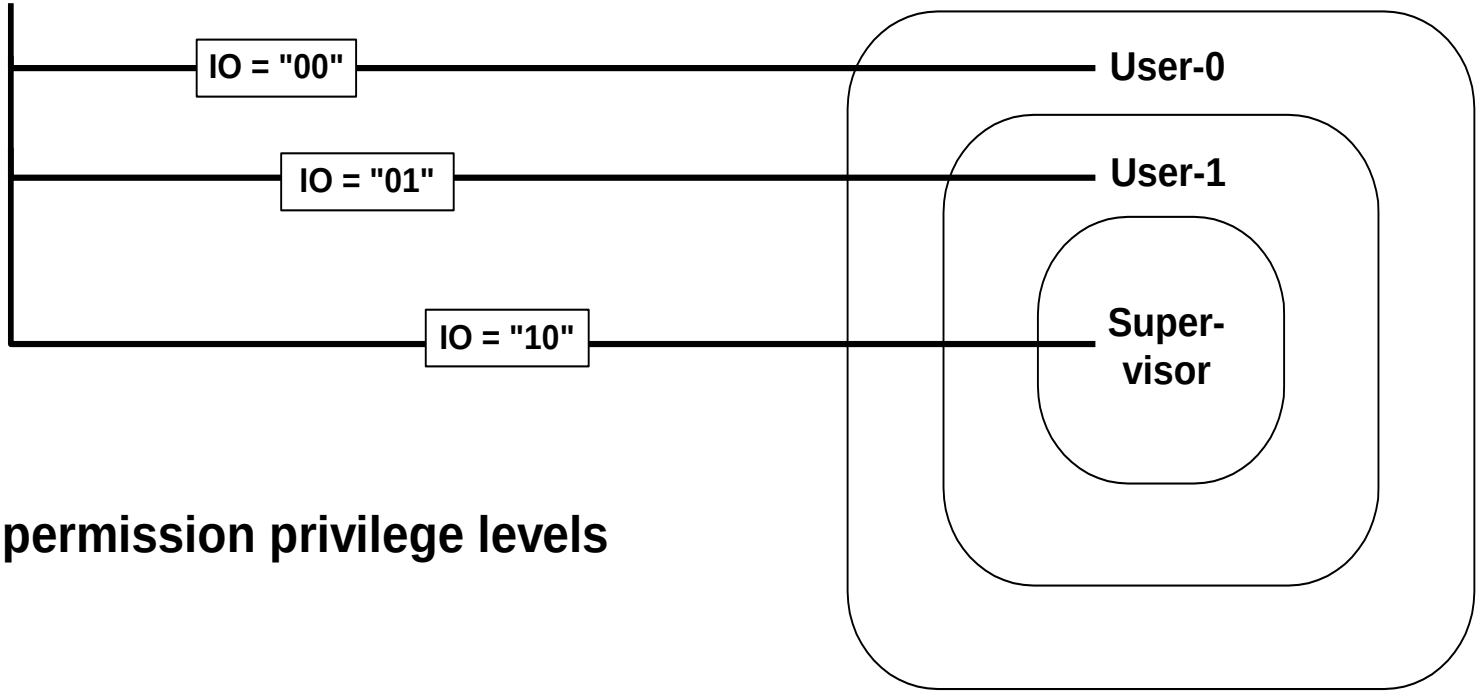
- TriCore protection system aims at :
  - Protecting core system functionality.
  - Protecting application against each other.
  - Providing test and debug functionality.

- TriCore protection system is based on:
  - Traps.
  - Permission levels.
  - Memory protection.

Permission Privilege Levels

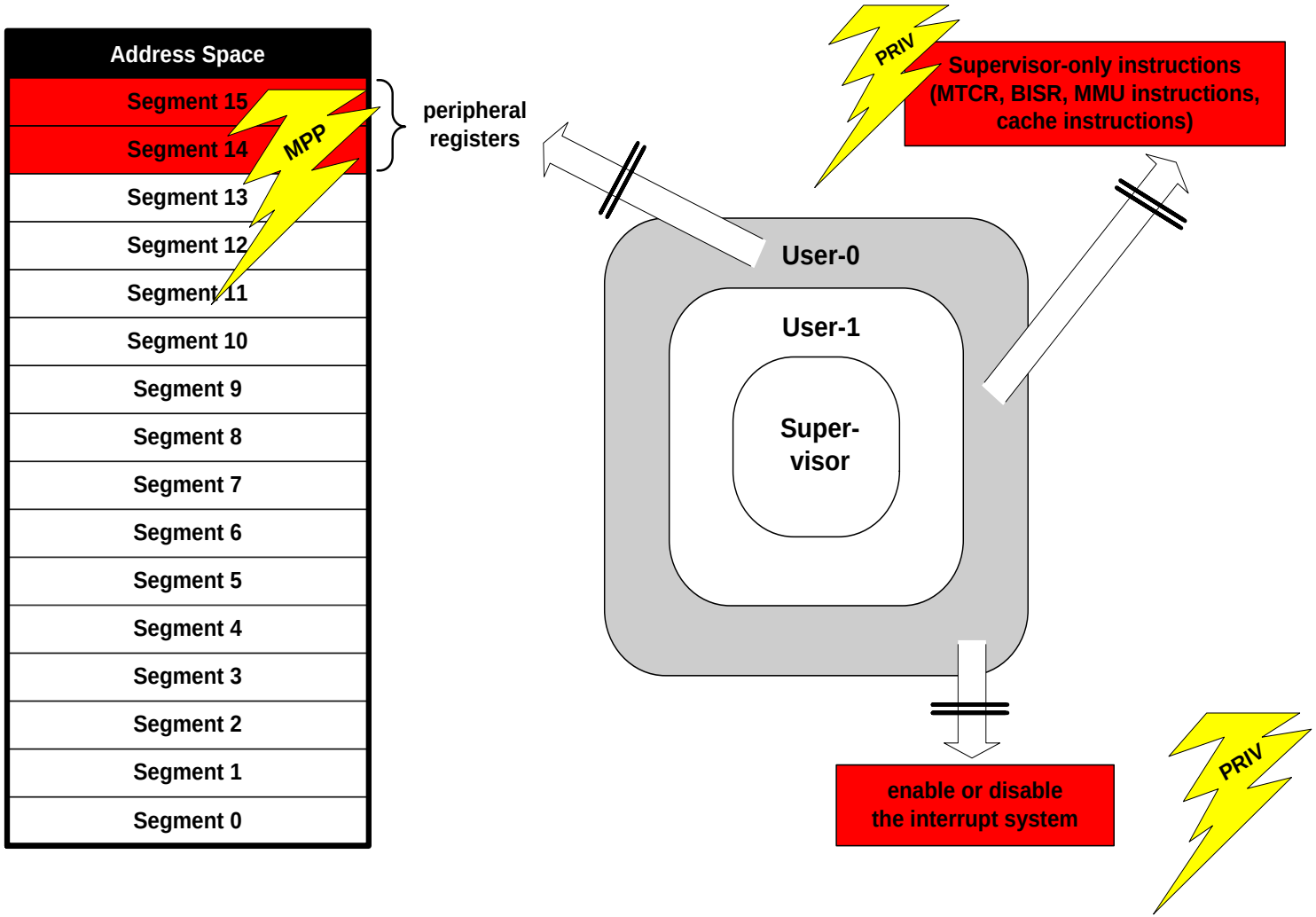
Program Status Word (PSW)



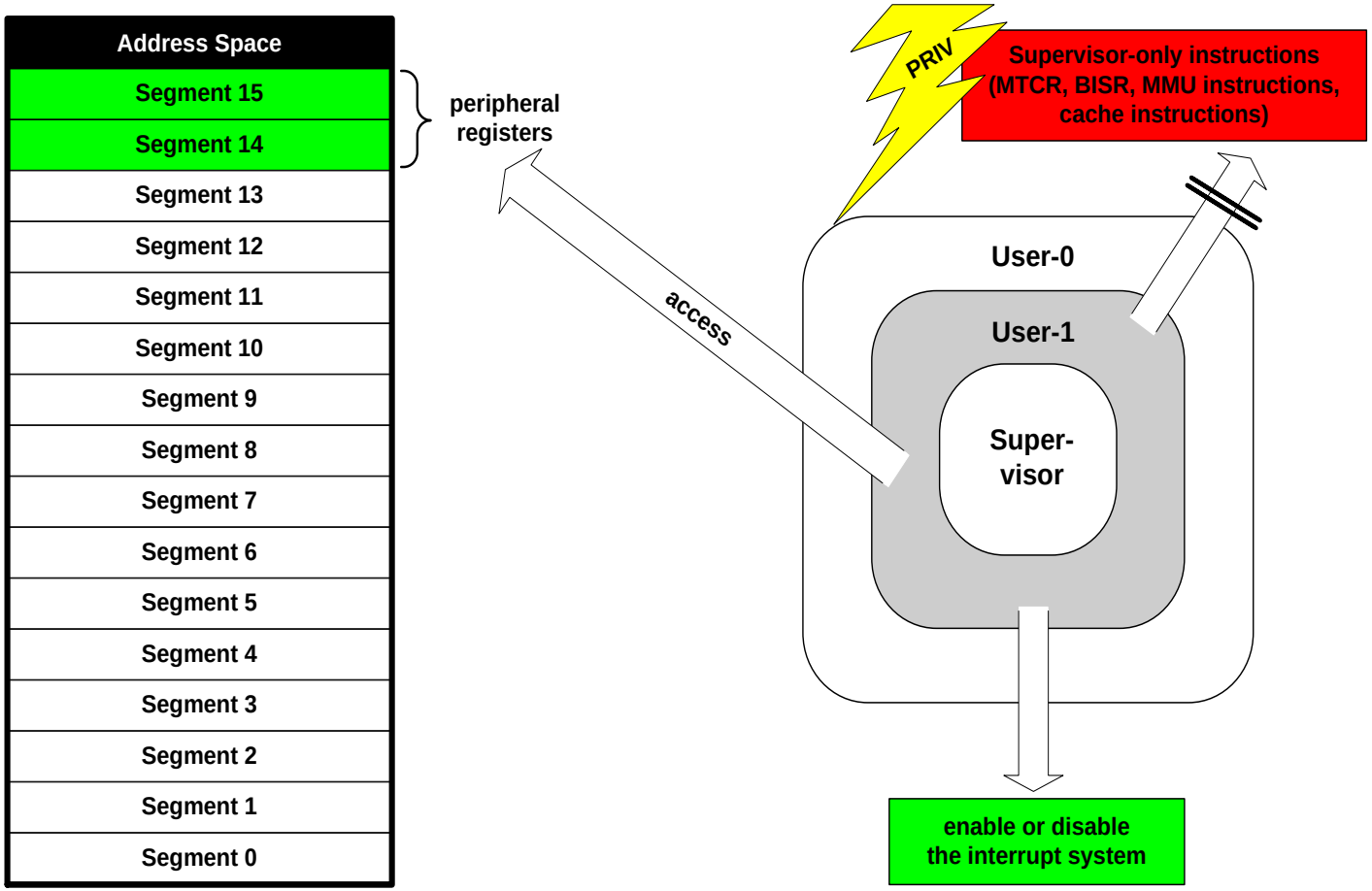


permission privilege levels

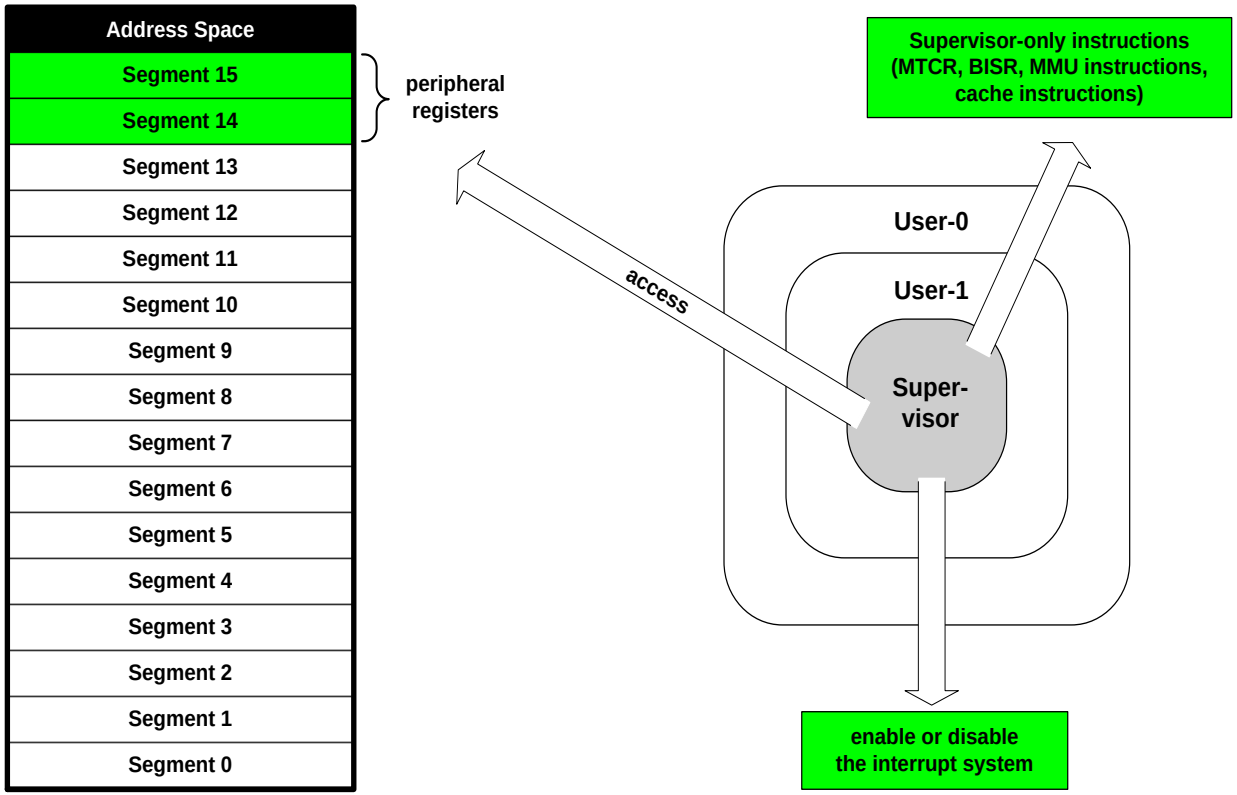
Permissions of User-0



Permissions of User-1



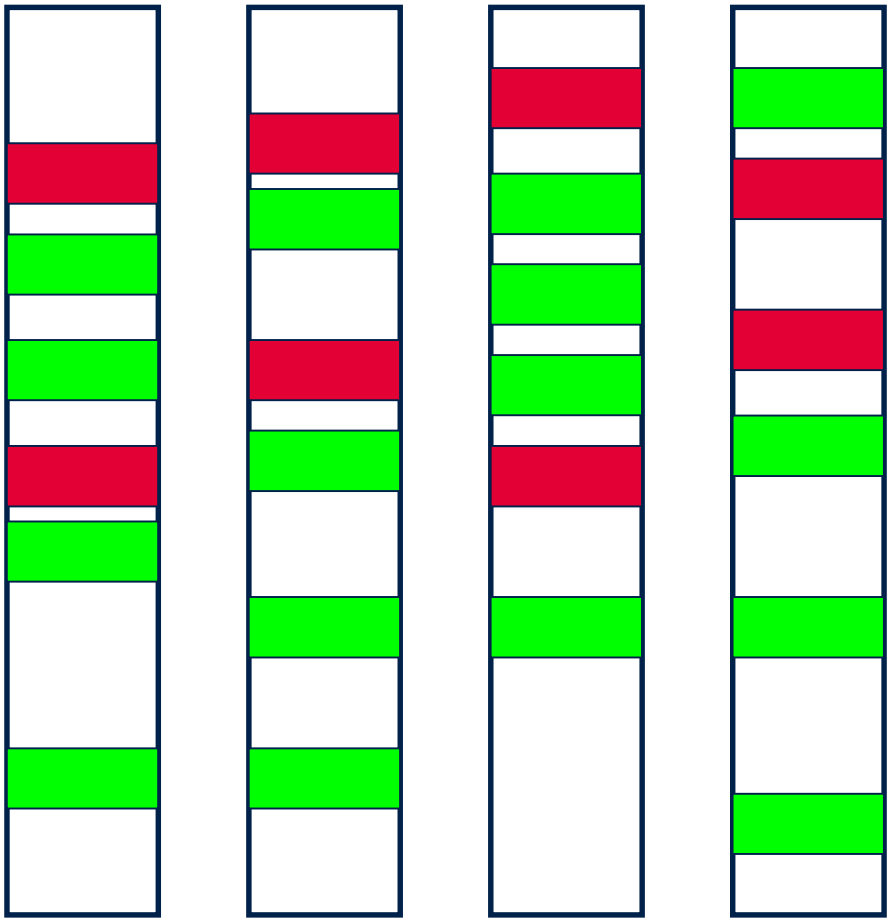
Permissions of Supervisor



3 TC系列MCU的CPU与外围电路

- （1）MCU基本架构、CPU与外围模块访问模式
- （2）CPU架构、主要寄存器及工作原理
- （3）MCU内存管理
  - Protection system
  - Address map
  - PMI/DMI/PMU
  - Flash/EEPROM

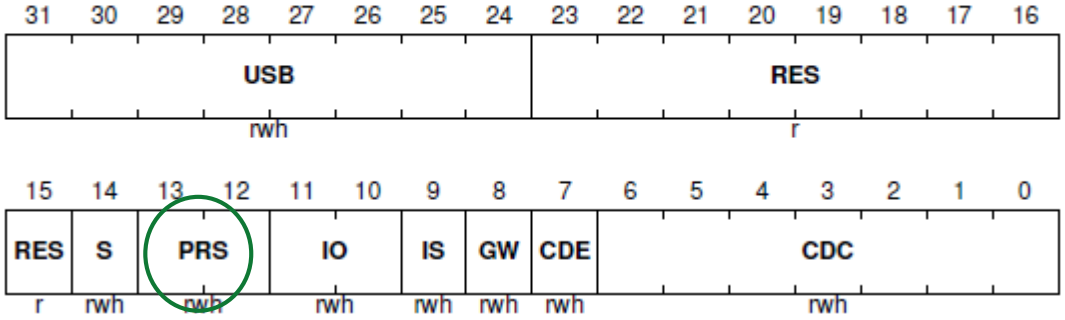
TriCore Memory Protection Overview



- Access permissions defined by address ranges
- Separate ranges dedicated for data access and program code fetch
- Data access permission defined by any combination of:
  - Read Enable
  - Write Enable
- Fetch Access permission
  - Access Enable
- Fast permission switching -> Protection Sets

TriCore Memory Protection CPU MPU Configuration

PSW
Program Status Word (F881 FE04$_H$) Reset Value: 0000 0B80$_H$

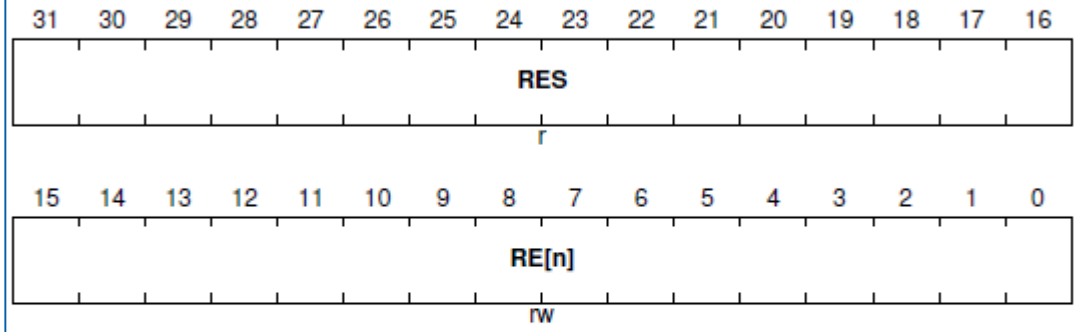


Field	Bits	Type	Description
PRS	[13:12]	rwh	**Protection Register Set** Selects the active Data and Code Memory Protection Register Set. The memory protection register values control load, store and instruction fetches within the current process. 00_B Protection Register Set 0. 01_B Protection Register Set 1. 10_B Protection Register Set 2. 11_B Protection Register Set 3.
S	14	rwh	**Safe Task Identifier**

- The PSW.PRS is used to select an Protection Register Set for the task to be executed

TriCore Memory Protection CPU MPU Configuration

DPRE_x (x=0-3)
Data Protection Read Enable Set Configuration Register x
($E010_H+x*4_H$)
Reset Value: Implementation Specific

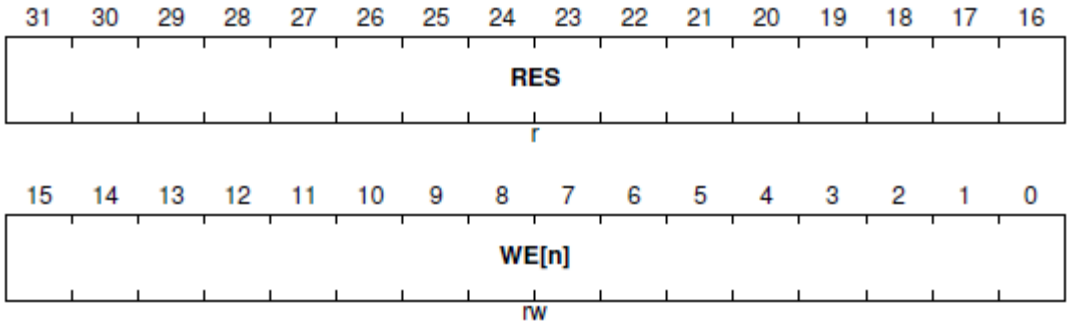


Field	Bits	Type	Description
RE[n]	[15:0]	rw	**Data protection Range Read Enable** 0 : Data read accesses to data protection range[n] not permitted. 1 : Data read accesses to data protection range[n] permitted.
RES	[31:0]	r	**Reserved**

- 4 Data Protection Register sets selected by PSW.PRS[1:0]
  - DPRE_x allows to specify read access to any of the 16 MPU regions

TriCore Memory Protection CPU MPU Configuration

DPWE_x (x=0-3)
Data Protection Write Enable Set Configuration Register x
($E020_H+x*4_H$)
Reset Value: Implementation Specific

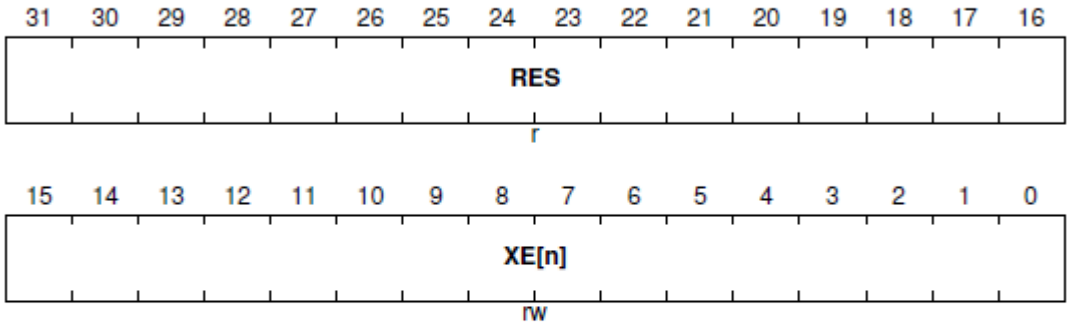


Field	Bits	Type	Description
WE[n]	[15:0]	rw	Data protection Range Write Enable 0 : Data write accesses to data protection range[n] not permitted. 1 : Data write accesses to data protection range[n] permitted.
RES	[31:0]	r	Reserved

- 4 Data Protection Register sets selected by PSW.PRS[1:0]
  - DPWE_x allows to specify write access to any of the 16 MPU regions

TriCore Memory Protection CPU MPU Configuration

CPXE_x (x=0-3)
Code Protection Execute Enable Set Configuration Register x
($E000_H+x*4_H$)
Reset Value: Implementation Specific



Field	Bits	Type	Description
XE[n]	[15:0]	rw	**Code protection Range Execute Enable** 0 : Execute accesses to code protection range[n] not permitted. 1 : Execute accesses to code protection range[n] permitted.
RES	[31:16]	r	**Reserved**

- 4 Code Protection Register sets selected by PSW.PRS[1:0]
  - CPXE_x allows to specify read access to any of the 8 code MPU regions

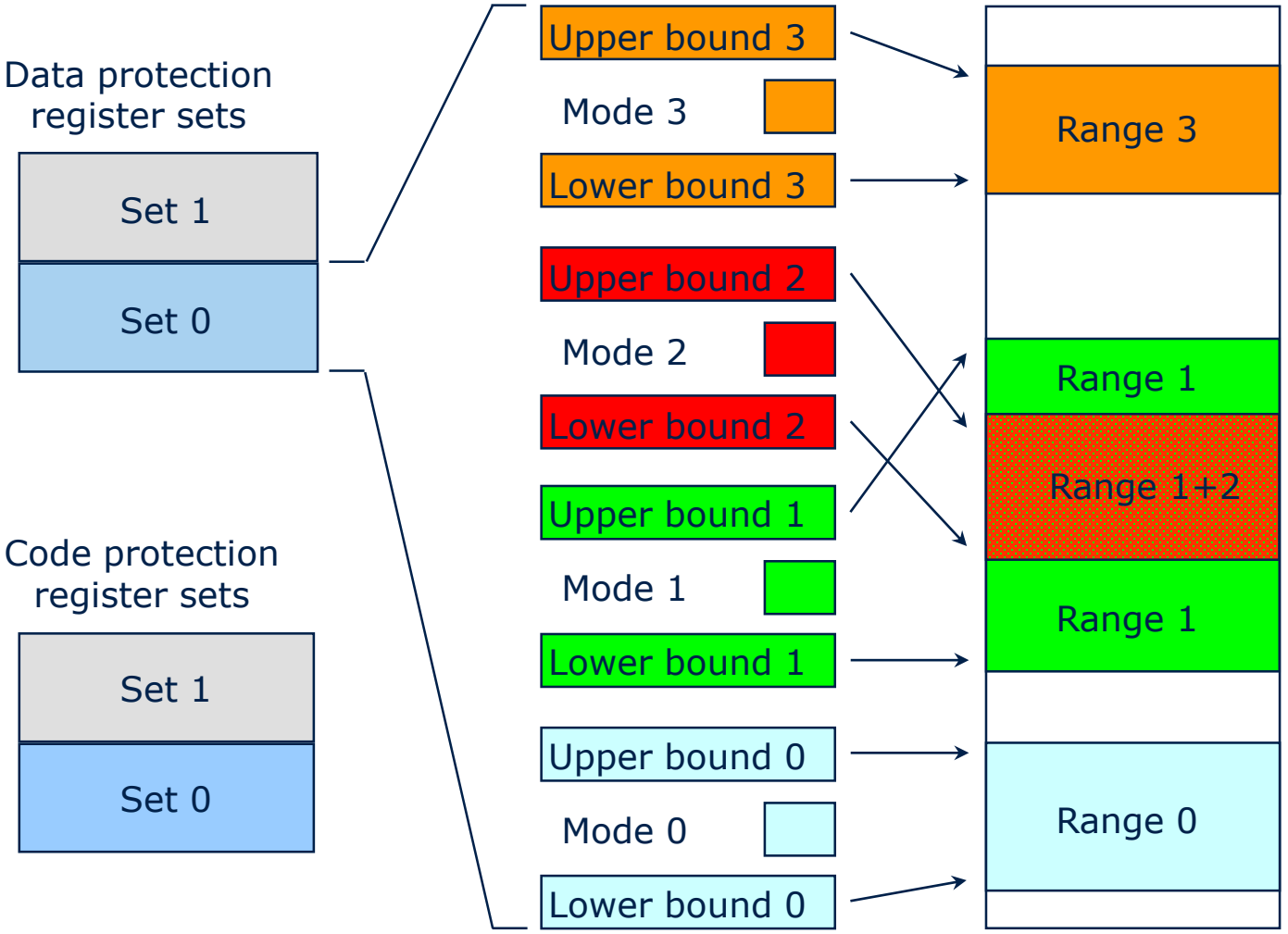
Memory Protection System

- Each set of register defined several data / code ranges.

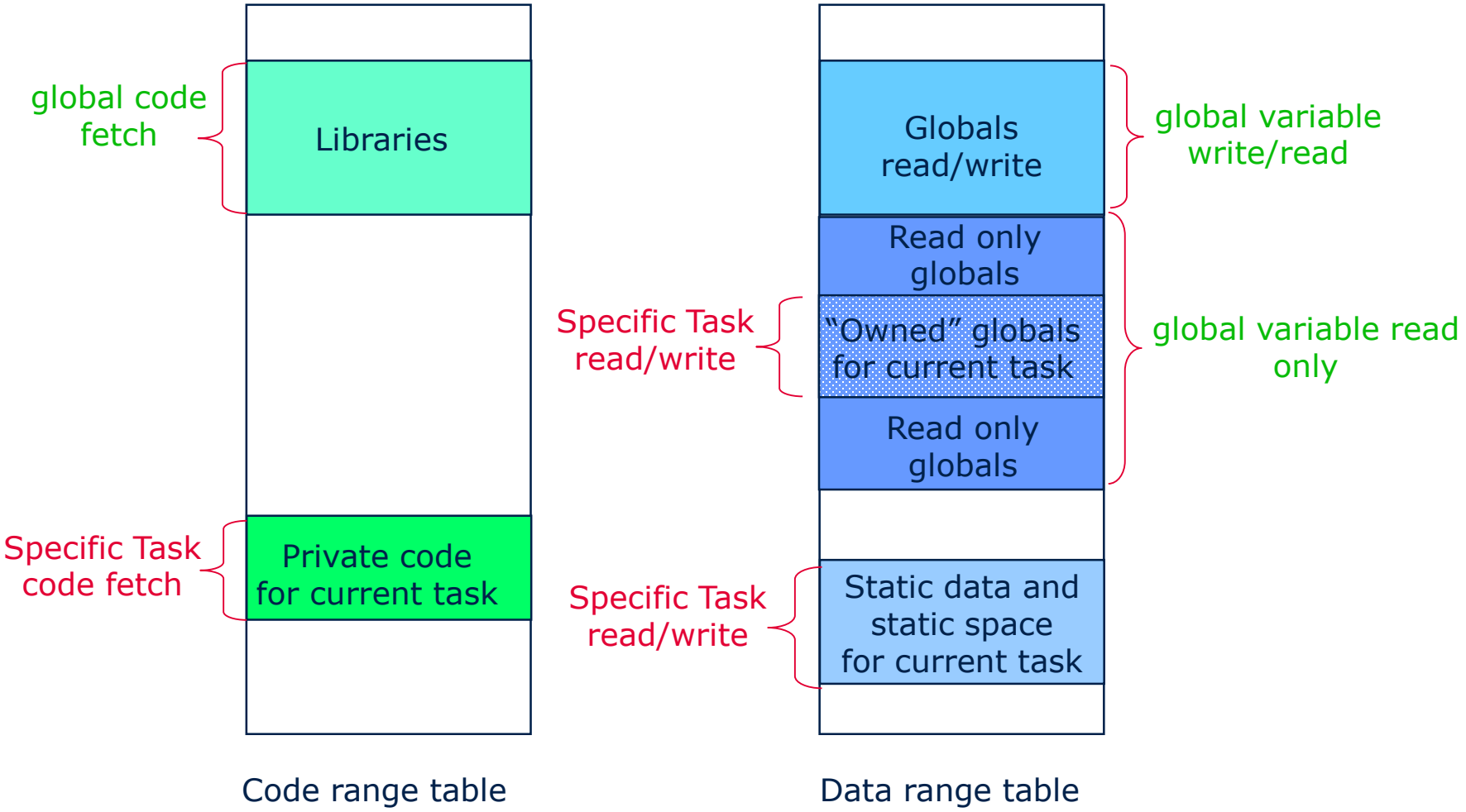
- Each range is defined by :
  - An upper boundary register.
  - An lower boundary register.
  - A permission mode.

- Permission mode:
  - Data Range: Read enable, write enable (or both)
  - Code Range: Instruction fetch enable.

Example data memory ranges.

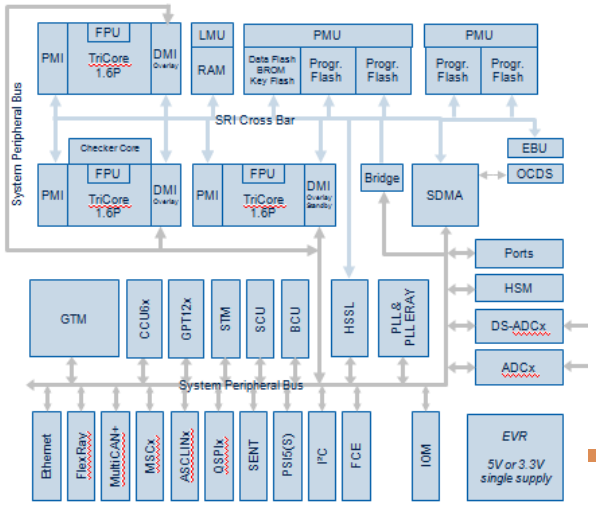


Example code and data memory ranges



Aurix Address Map

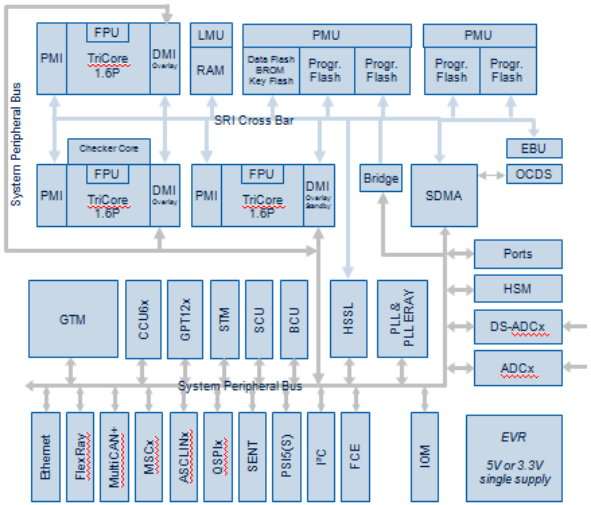
- TriCore 4GB Addressable **Memory Map** is organized into Segment of 256 MBytes.
- A Segment is identified by the A[31:28] bits of the system address
- Each Segment allows access to specific area location. Also used to define cacheable **(可缓存的)** and non-cacheable areas
- Segment C and D is now used as private views by it own CPU.



- **Segments 0-7:** Access to the CPUs Program and Data Scratch Pad SRAM
- **Segment 8:** Cached segment: Access to PFlash, Boot ROM and EBU (if implemented)
- **Segment 9:** Access to LMU SRAM (if implemented) and EMEM (Emulation Device only)
- **Segment 10:** Non-cached segment: access to PFlash, DFlash, Boot ROM and EBU (if implemented)
- **Segment 11:** Access to LMU SRAM (if implemented) and EMEM (Emulation Device only)
- **Segment 12-13:** These memory segment are reserved
- **Segment 14:** Reserved
- **Segment 15:** Accesses to all Module Registers (and local SRAMs) and CPU Special Registers. No User0 access

Aurix Multiprocessor Address Map

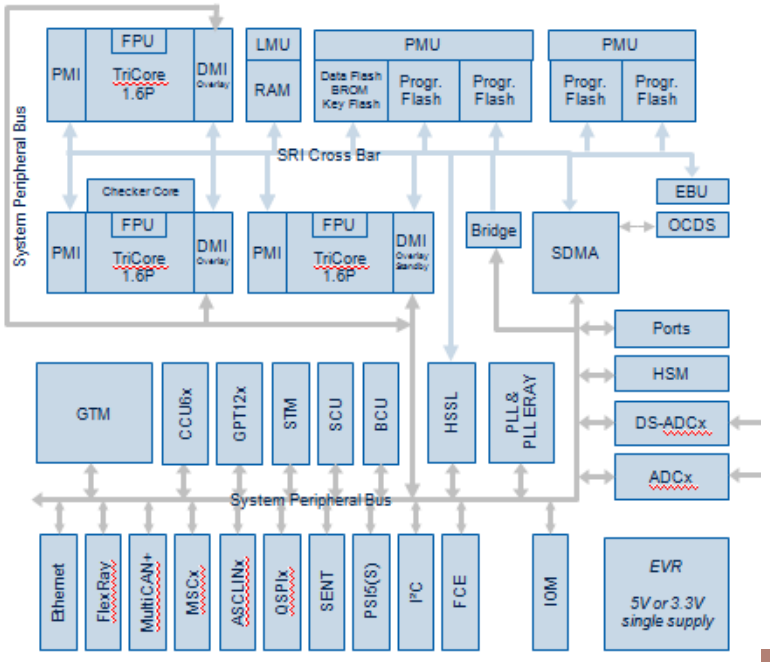
- The Aurix Multiprocessor address map supports up to eight processors
- Each CPU has access to its own
  - Code Scratch memory at local address **0xC000 0000**
  - Data Scratch memory at local address **0xD000 0000**
- Each CPU has also access to its own memory via its global segment (0-7), Data Scratch at offset **0** and Code Scratch Pad Memory at offset **0x0010 0000 (1MB)**



Segment Number	
0xD	CPU0..7 Data Scratch memory
0xC	CPU0..7 Code Scratch memory
0x7	CPU0 Code+Data Scratch
0x6	CPU1 Code+Data Scratch
0x5	CPU2 Code+Data Scratch
0x4	CPU3 Code+Data Scratch
0x3	CPU4 Code+Data Scratch
0x2	CPU5 Code+Data Scratch
0x1	CPU6 Code+Data Scratch
0x0	CPU7 Code+Data Scratch

Aurix Multiprocessor Address Map – Local Memories

- Data access to own Data Scratch or Instruction fetch from own Instruction Scratch is always non-cached
- Access to memories of other CPUs can be cached or non-cached (control per CPU via Memory Attribute).
- Physical Memory Attribute Registers (PMA) control the cached/non-cached behaviour of Code Fetch and Load/Store accesses (individually) to non-local CPU memories
- All SRAMs can be used as Overlay SRAM:
  any DSPR SRAM, EMEM, LMU SRAM, external memory



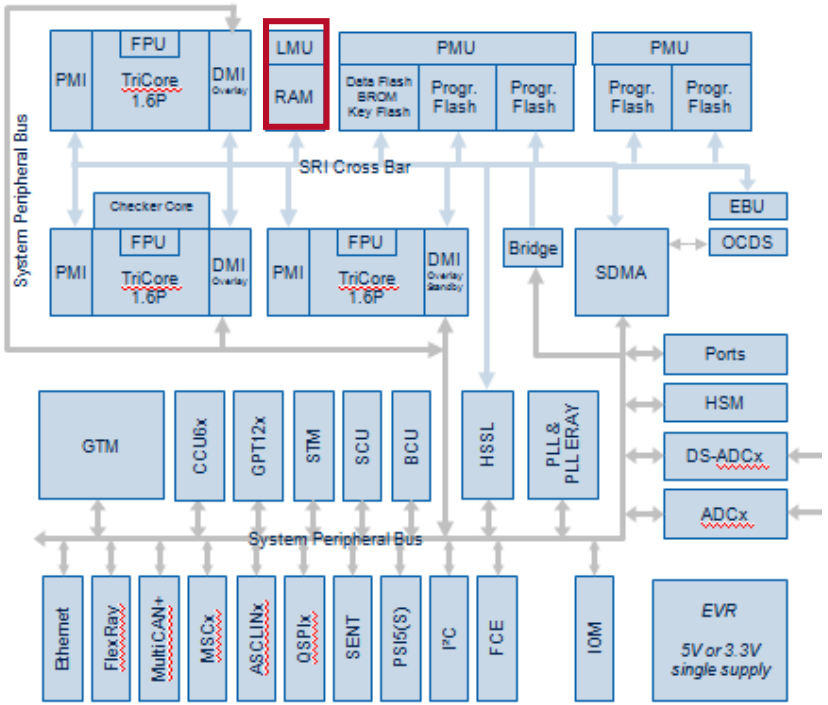
Segment Number	
0xD	CPU0..7 Data Scratch memory
0xC	CPU0..7 Code Scratch memory
0x7	CPU0 Code+Data Scratch
0x6	CPU1 Code+Data Scratch
0x5	CPU2 Code+Data Scratch
0x4	CPU3 Code+Data Scratch
0x3	CPU4 Code+Data Scratch
0x2	CPU5 Code+Data Scratch
0x1	CPU6 Code+Data Scratch
0x0	CPU7 Code+Data Scratch

Aurix Multiprocessor Address Map – LMU and EMEM Memory

- LMU **(Local Memory Unit)** Memory / EMEM

9	$9000\ 0000_H$ - $9000\ 7FFF_H$	32 Kbyte	LMU SRAM (LMURAM)	access	access
	$9000\ 8000_H$ - $9EFF\ FFFF_H$	-	Reserved	SRIBE	SRIBE
	$9F00\ 0000_H$ - $9F0F\ FFFF_H$	1 Mbyte	Reserved for TC27x Emulation Device Memory (EMEM)	SRIBE	SRIBE
	$9F10\ 0000_H$ - $9FFF\ FFFF_H$	-	Reserved	SRIBE	SRIBE

11	$B000\ 0000_H$ - $B000\ 7FFF_H$	32 Kbyte	LMU SRAM (LMURAM)	access	access
	$B000\ 8000_H$ - $BEFF\ FFFF_H$	-	Reserved	SRIBE	SRIBE
	$BF00\ 0000_H$ - $BF0F\ FFFF_H$	1 MB	Reserved for TC27x Emulation Device Memory (EMEM)	SRIBE	SRIBE
	$BF10\ 0000_H$ - $BFFF\ FFFF_H$	-	Reserved	SRIBE	SRIBE



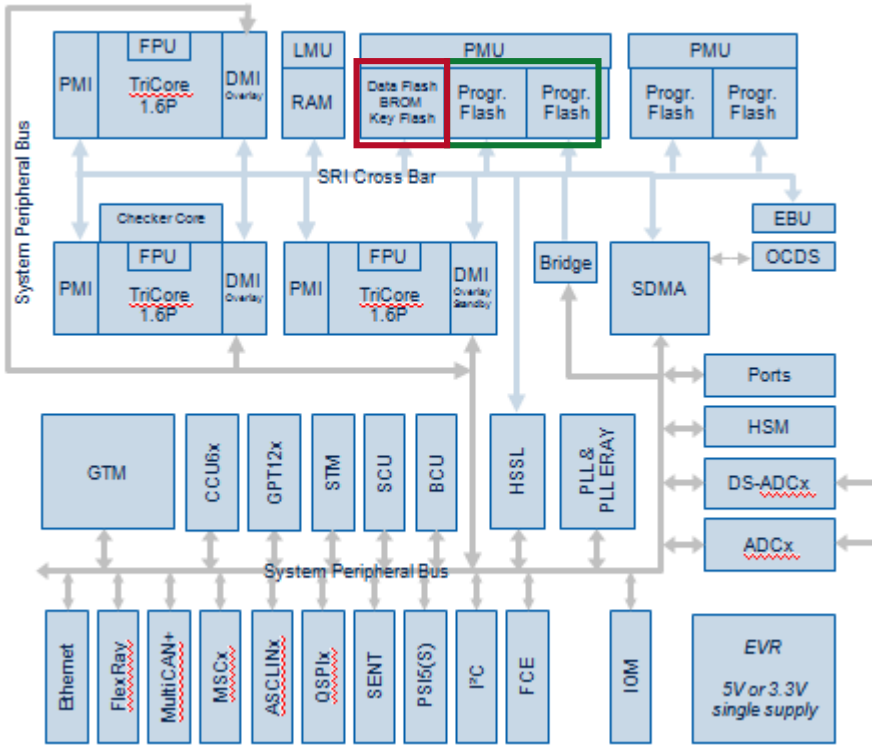
Aurix Multiprocessor Address Map – Flash

- Program Flash
- Pflash cached in segment 8

8	$8000\ 0000_H$ - $801F\ FFFF_H$	2 Mbyte	Program Flash 0 (PF0)
	$8020\ 0000_H$ - $803F\ FFFF_H$	2 Mbyte	Program Flash 1 (PF1)
	$8040\ 0000_H$ - $8FE6\ FFFF_H$	-	Reserved

- Data Flash only non-cached
- Pflash / Dflash in segment A

10	$A000\ 0000_H$ - $A01F\ FFFF_H$	2 Mbyte	Program Flash 0 (PF0)
	$A020\ 0000_H$ - $A03F\ FFFF$	2 Mbyte	Program Flash 1 (PF1)
	$A040\ 0000_H$ - $AEFF\ FFFF_H$	-	Reserved
	$AF00\ 0000_H$ - $AF0F\ FFFF_H$	1 Mbyte	Data Flash 0 (DF0)
	$AF10\ 0000_H$ - $AF10\ 3FFF_H$	16 Kbyte	Data Flash 0 (DF0)
	$AF10\ 4000_H$ - $AF10\ FFFF_H$	-	Reserved
	$AF11\ 0000_H$ - $AF11\ FFFF_H$	64 Kbyte	Data Flash 1 (DF1)
	$AF12\ 0000_H$ - $AFE6\ FFFF_H$	-	Reserved



RAM: Random Access Memory 放变量
ROM: Read only Memory 放程序
EPROM: Electrical Program Read only Memory

EEPROM: Erasable Electrical Program Read only Memory

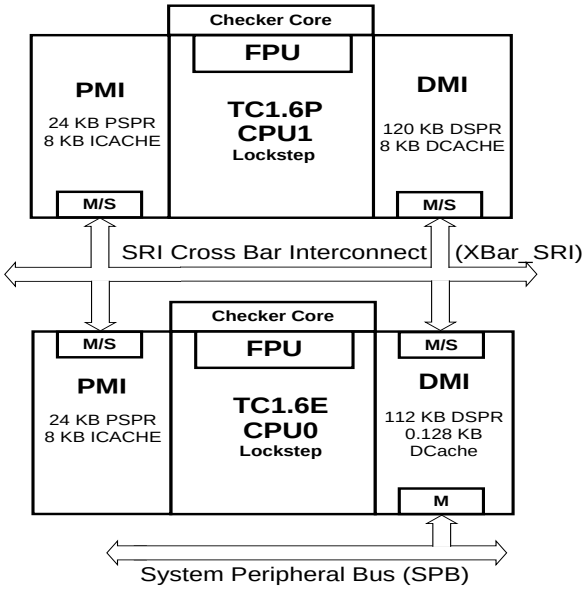
TC1.6P/E: PMI/DMI Overview

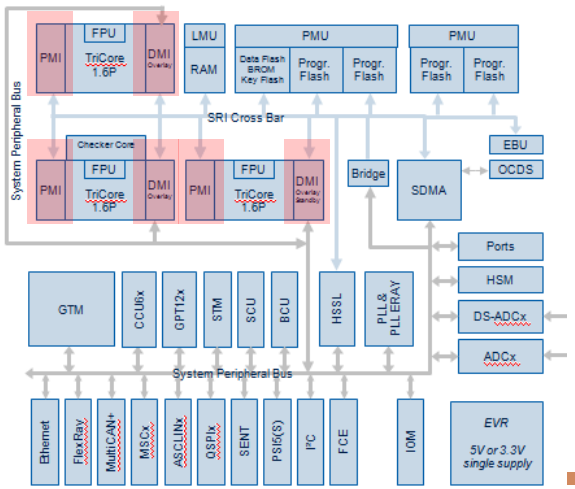
Program Memory Interface (PMI)

- Program Scratch Pad SRAM (PSPR)
- Program Cache SRAM (PCache)
- Program Cache TAG SRAM (PTAG)

Data Memory Interface (DMI)

- Data Scratch Pad SRAM (DSPR)
- Data Cache SRAM (DCache)
- Data Cache TAG SRAM (DTAG)

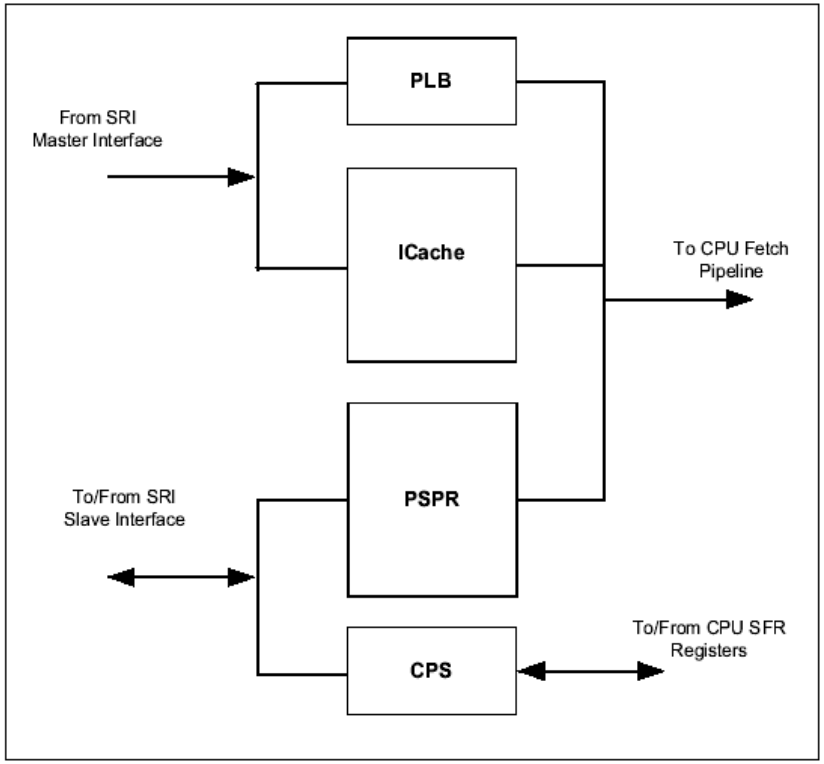




7	7000 0000$_H$ - 7001 BFFF$_H$	112 Kbyte	CPU0 Data Scratch-Pad SRAM (CPU0.DSPR)	access	access
	7001 C000$_H$ - 700F FFFF$_H$	-	Reserved	SRIBE	SRIBE
	7010 0000$_H$ - 7010 5FFF$_H$	24 Kbyte	CPU0 Program Scratch-Pad SRAM (CPU0.PSPR)	access	access
	7010 6000$_H$ - 7017 FFFF$_H$	-	Reserved	SRIBE	SRIBE
	7018 0000$_H$ - 7018 1FFF$_H$	8 Kbyte	CPU0.Program Cache SRAM (CPU0.ICACHE)	access[3] / SRIBE	access[3] / SRIBE
	7018 2000$_H$ - 701B FFFF$_H$	-	Reserved	SRIBE	SRIBE
	701C 0000$_H$ - 701C 01FF$_H$	0.5 Kbyte	CPU0 Program Cache TAG SRAM[4] (CPU0.PTAG)	access[3] / SRIBE	access[3] / SRIBE
	701C 0200$_H$ - 7FFF FFFF$_H$	-	Reserved	SRIBE	SRIBE

TC1.6P/E: Program Memory Interface(PMI)

- **Two way set associative cache缓存**
  - Least-Recently Used replacement algorithm
  - Cache Line Size: 256 bits (4 double-words)
  - Program Line Buffer (PLB, 256 bits) for non cached fetches / bypass
  - Single cycle access for hit
  - Global invalidation and bypass support

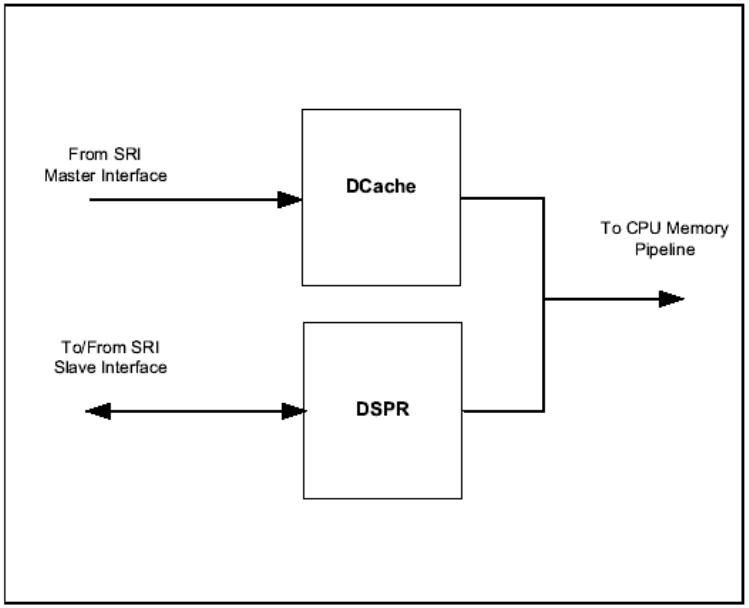


- **Program Scratch Pad (PSPR):**
  - Single cycle instruction fetch
  - Global resource via SRI

- **All PMI SRAMs are ECC protected**

TC1.6P: Data Memory Interface (DMI)

- **Two way set associative cache**
  - Least-Recently Used replacement algorithm
  - Cache Line Size: 256 bits (4 double-words)
  - Write Back Cache
  - Single cycle data access for hit
  - Single Cycle Invalidation (non-dirty cache lines)



- **Data Scratch Pad (DSPR):**
  - Single cycle data access
  - Supporting unaligned access (16 bit granularity)
  - Can be used as Data SRAM

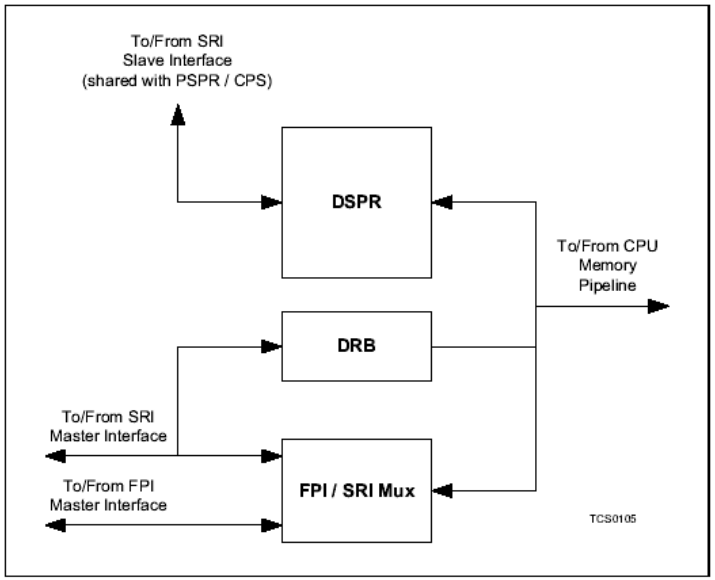
- **All DMI SRAMs are ECC protected**
- **Direct Master Interface to System Peripheral Bus**

TC1.6E: Data Memory Interface (DMI)

- **Four-entry fully associative Read Data Buffer**
  - Least-Recently Used replacement algorithm
  - Buffer Line Size: 256 bits
  - Single cycle data access for hit
  - Single Cycle Invalidation

- **Data Scratch Pad (DSPR):**
  - Single cycle data access
  - Supporting unaligned access (16 bit granularity)
  - Can be used as Data SRAM

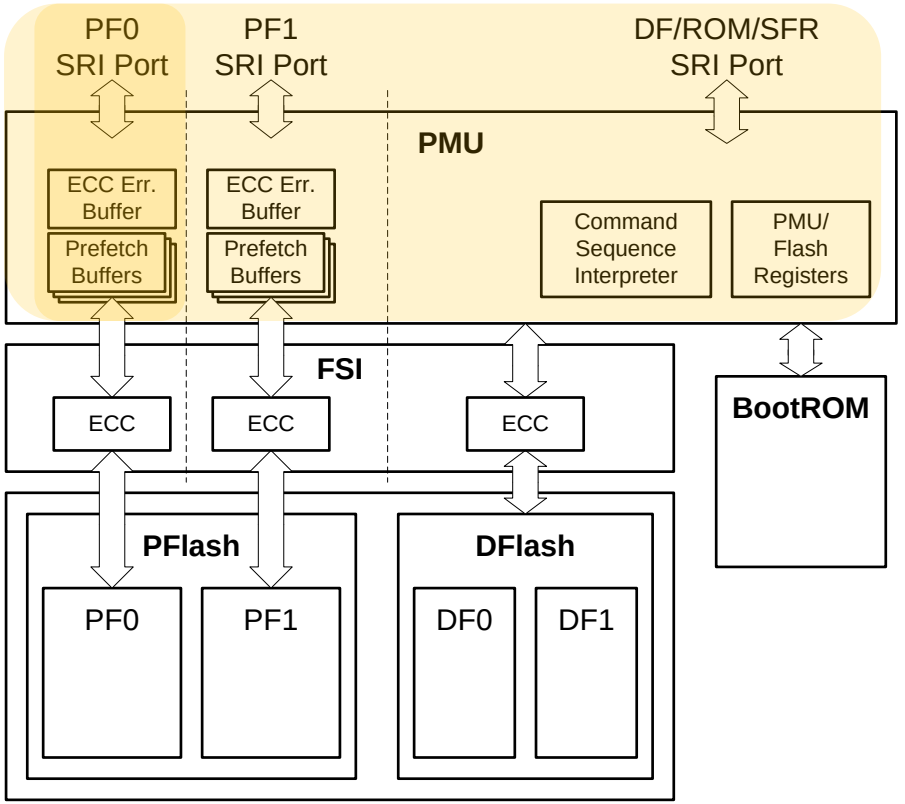
- **All DMI SRAMs are ECC protected**
- **Direct Master Interface to System Peripheral Bus**



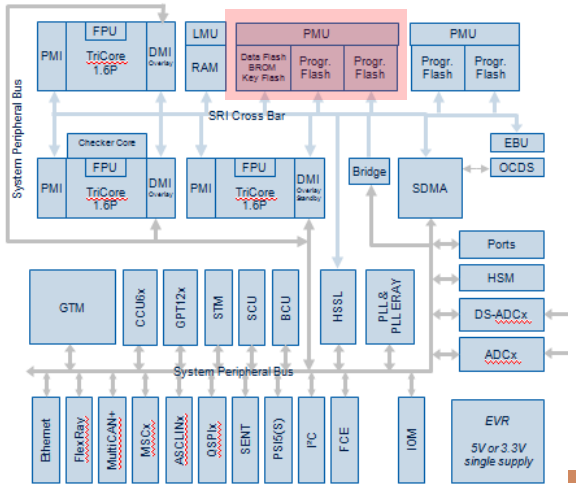
Program Memory Unite (PMU)
PFlash Access via XBAR_SRI

- PMU with multiple XBAR_SRI Slave Interfaces
  - Pflash0 module (TC27x: 2MB)
  - PFlash1 module (TC27x: 2MB)
  - DFlash modules , BootROM, PMU SFR

- Each PFlash Slave Interface:
  - Support of pipelined access via XBAR_SRI
  - One independent Read Path
  - Three PMU 256-bit Read/Pre-fetch buffer
  - Automatic Pre-fetch of next 256 in the background

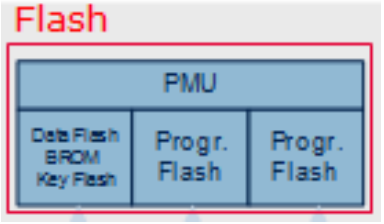


- Read While Write on PFlash /DFlash



Flash

- Flash concept changed from Fowler-Nordheim (AUDO MAX) to HS3P (AURIX) Hot-Electron for Programming sequences



- Motivation
  - Today the **end-of-line programming time** is limited by the speed of CAN. With the availability of FlexRay and Ethernet this bottleneck is removed.
  - The new bottleneck is **the flash erasing and programming time** due to Infineon´s current flash cell concept.
  - With increasing flash sizes this disadvantage gets more and more critical.
  - The new flash concept significantly increase the flash erase and programming speed in the range of **10-20 times faster** than today (e.g. for 5V burst programming up to 1,3 MByte/s)

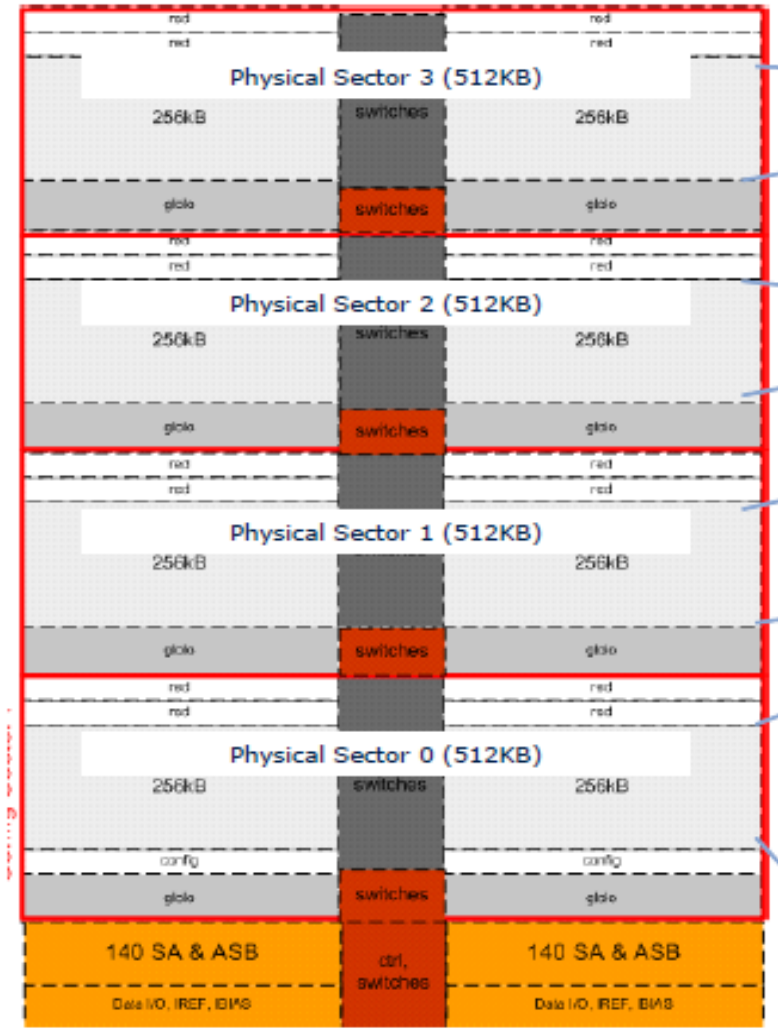
Fast Erase times (1MB: HS3P (65nm): <2s, FN (90nm) ~ 20s)
Fast Programming (1MB: HS3P (65nm): ~1s , FN (90nm) ~ 20s)

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Aurix- Program Flash – Program Time

- PFlash Prog Page (32 Bytes):
  - 5V Operation: 70µs
  - 3.3V Operation: 110µs
- PFlash Prog Burst (256 Bytes):
  - 5V Operation: 170µs
  - 3.3V Operation: 470µs
- => Pflash Prog Throughput (5V Operation):
  - page mode: ~2.5s/1MB; ~ 19s/8MB or **0,4MB/s**
  - burst mode with 256Byte buffer: <0,8s/1MB; <6s for 8MB or **1,3MB/s**
- VDDP current consumed by programming is higher (~10mA) for 3.3V operation.
- Independent of 5V/3.3V:
  - DFlash programming times, Erase times, Read times

Sectorization for 2MB Program Flash



Sub - Sector	Size [KB]
26	256
25	256

Sub - Sector	Size [KB]
24	256
23	256

Sub - Sector	Size [KB]
22	128
21	128
20	128
19	64
18	64

Sub - Sector	Size [KB]
17	64
16	64
15	32
14	32
13	32
12	32
11	32
10	32
9	32
8	32
7	16
6	16
5	16
4	16
3	16
2	16
1	16
0	16

AURIX - EEPROM Overview
(timings are target values, final values may be different)

- Size:
  - Physical Size = ***Factor*** x EEPROM (Logical) Size
  - ***Factor*** = 6 assuming 500.000 EEPROM updates + some overhead for emulation
- Endurance = 125.000
- Program:
  - Prog Granularity: 8 Byte (in addition a burst mode of 32byte is available)
  - Prog Time per page (8 Byte) ~ 70us
  - Prog Time per burst (32 Byte) ~ 140us
- Erase:
  - Erase Granularity: Sector (64KB)
  - Erase Time (Sector) ~ 1sec
- Read Access Time: < 100ns (target, final value after characterization)

小结

- 理解TC21x单核单片机内部架构，XBAR_SRI和SPB
- 理解SRI Master-slave及优先级仲裁功能
- 理解TC27x三核单片机内部架构，XBAR_SRI和SPB
- 理解Four-stage Super-scalar的Pipeline程序流程
- 理解代码运行模式User0、User1、Supervisor的差异
- 理解PSW.PRS与DPRE、DPWE、CPXE等单片机联合起来定义地址空间的方法
- 理解TriCore单片机地址空间的16个segments的基本功能定义
- 了解Flash、RAM、ROM和EEPROM的功能差异

本节结束，谢谢！

Thank you！